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Antoniolli et al.

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(54) **DEPOSITION APPARATUS, METHOD OF DEPOSITION ON A SUBSTRATE, SUBSTRATE STRUCTURE AND SUBSTRATE SUPPORT**

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See application file for complete search history.

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Primary Examiner — Michael G Miller

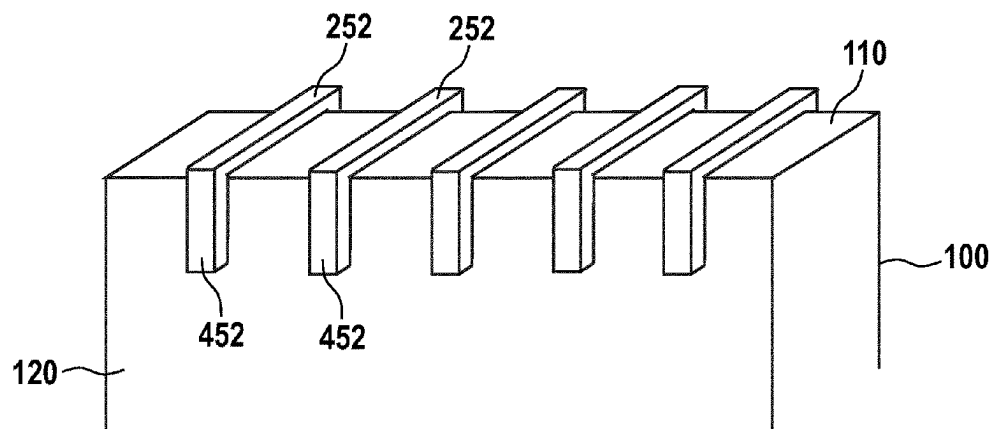
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(57) **ABSTRACT**

A deposition apparatus includes a first substrate support for supporting a substrate in a substantially vertical orientation. The substrate has a first main surface, a second main surface opposite the first main surface and a side surface between the first main surface and the second main surface. The deposition apparatus includes a first deposition device for depositing a first conductive pattern or a first resist mask on the side surface of the substrate while the substrate is supported in the substantially vertical orientation by the first substrate support.

9 Claims, 14 Drawing Sheets

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Fig. 1

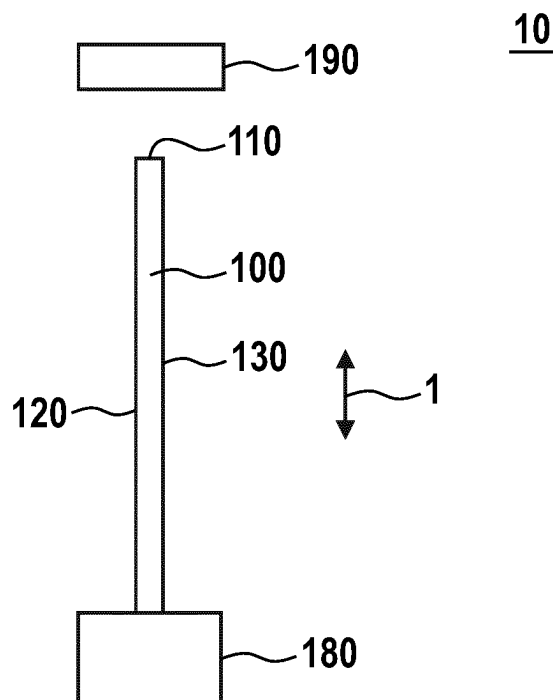


Fig. 2a

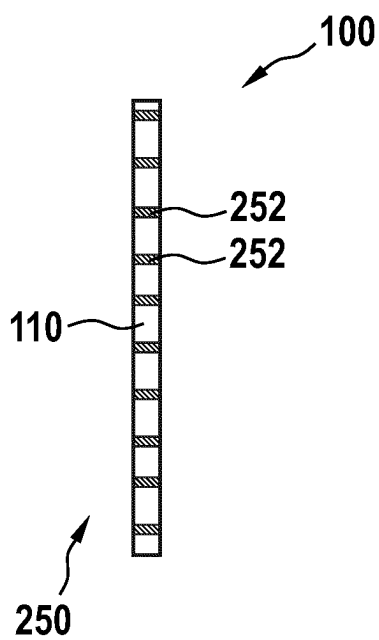


Fig. 2b

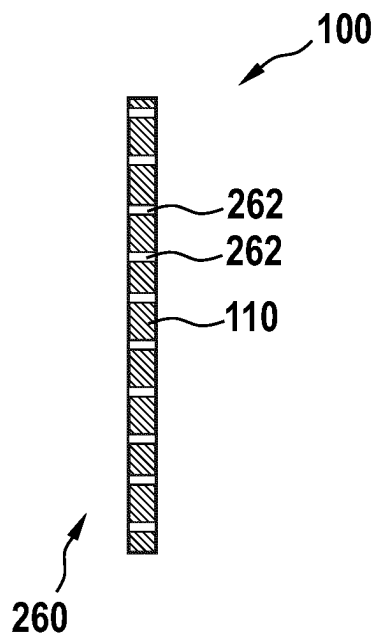


Fig. 3a



Fig. 3b

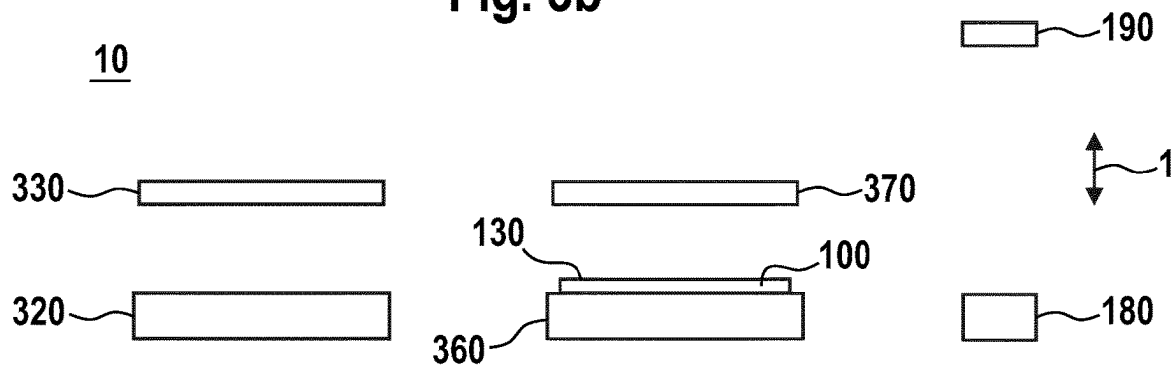


Fig. 3c

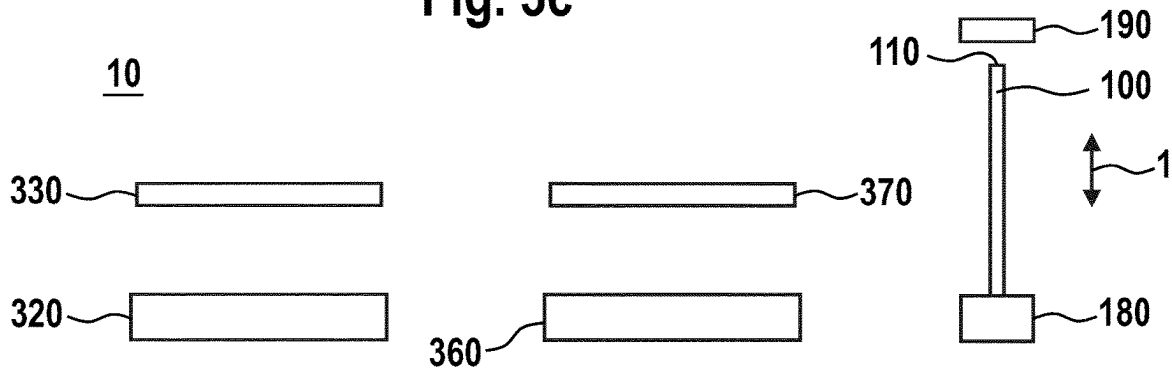


Fig. 4a

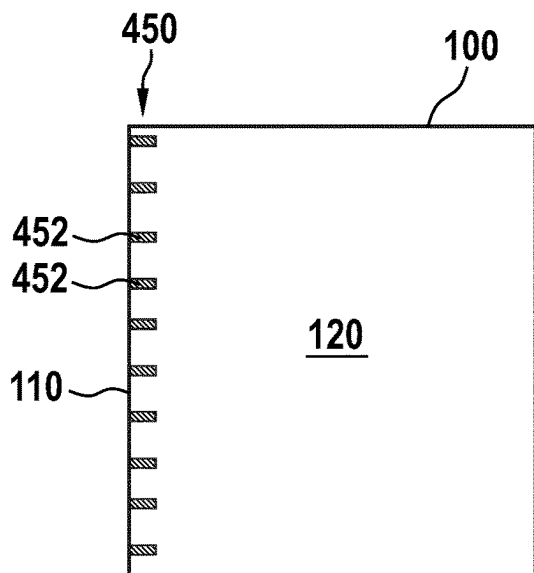


Fig. 4b

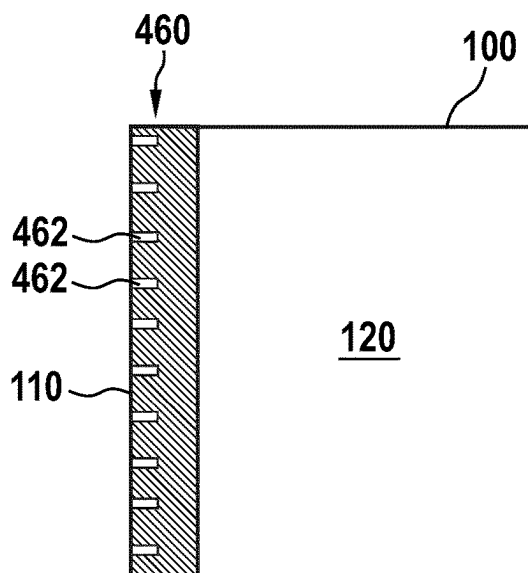


Fig. 5a

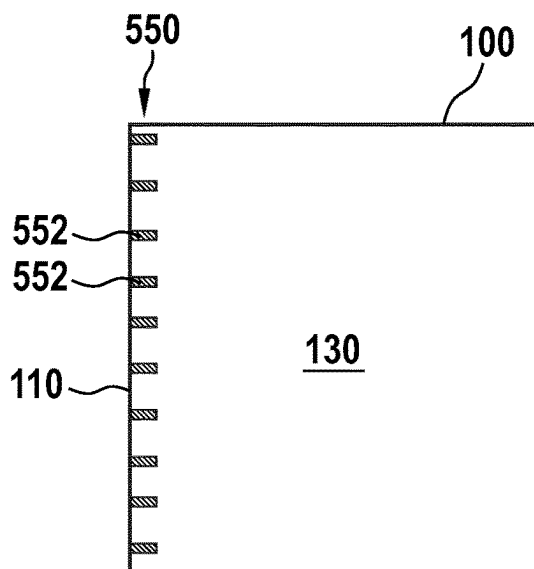


Fig. 5b

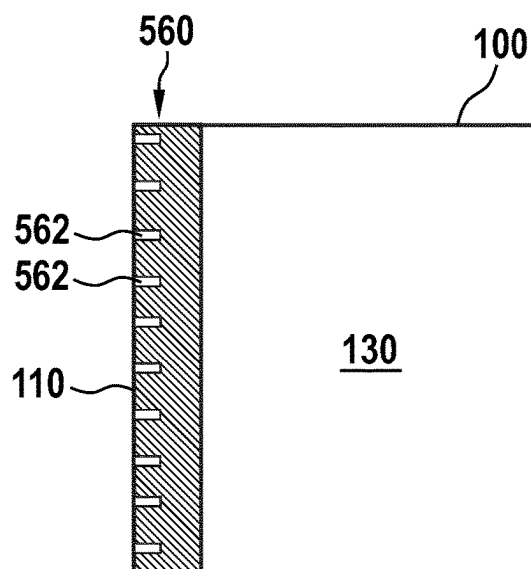


Fig. 6

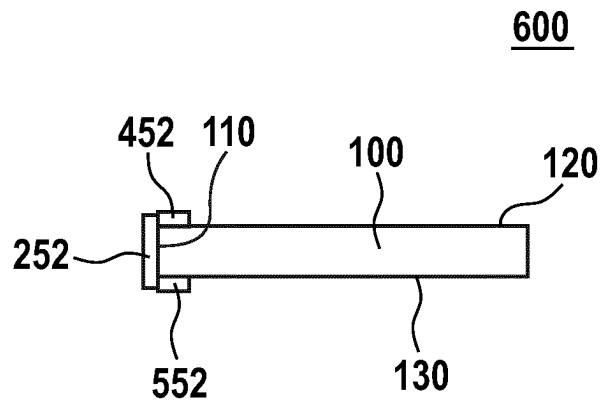


Fig. 7

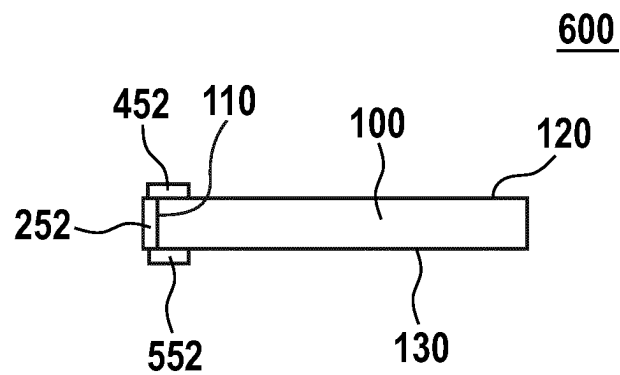


Fig. 8

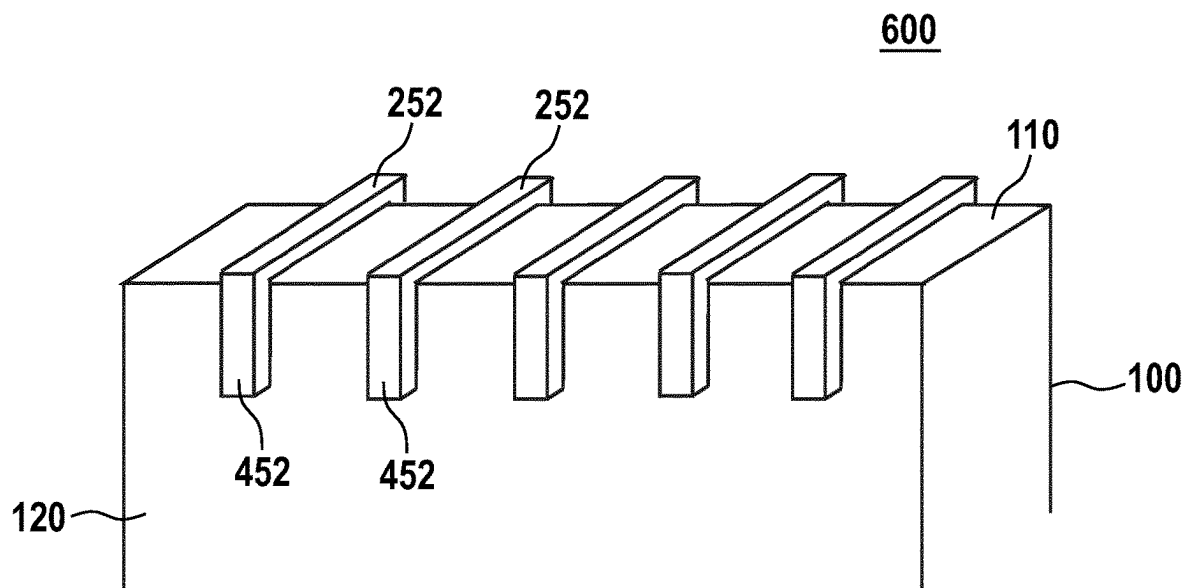


Fig. 9

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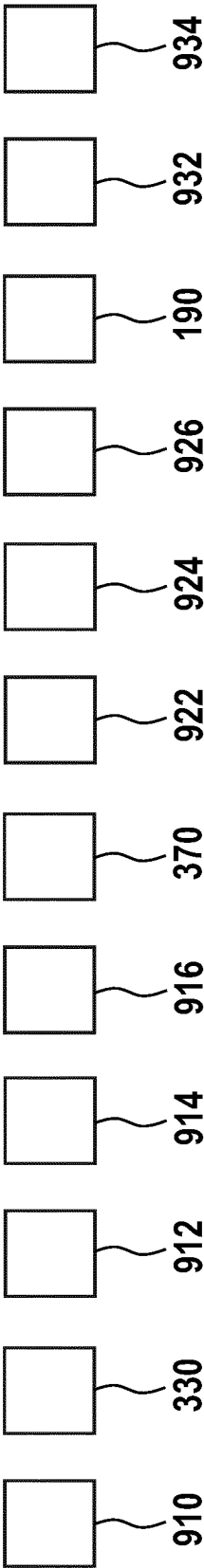


Fig. 10

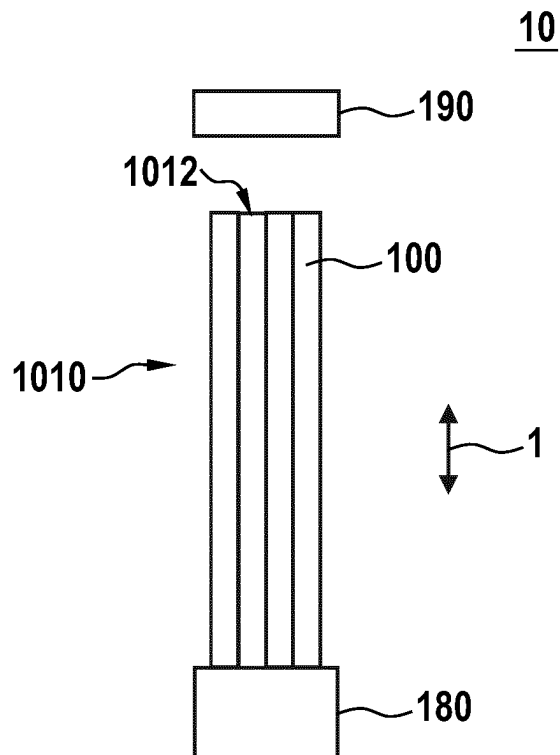


Fig. 11a

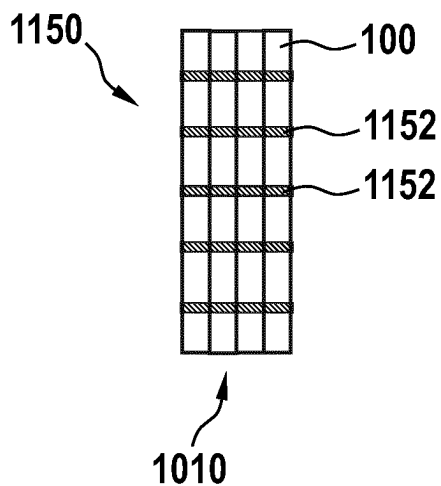


Fig. 11b

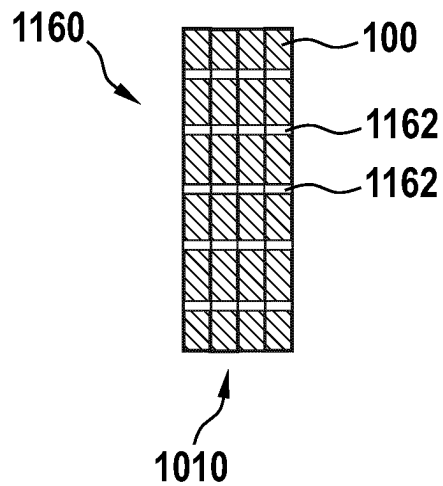


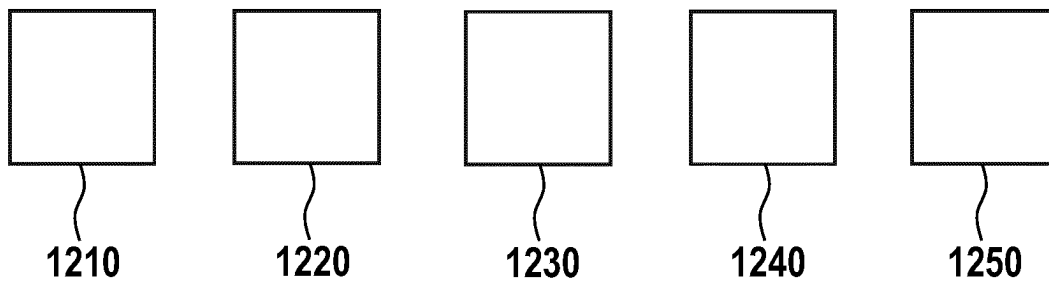
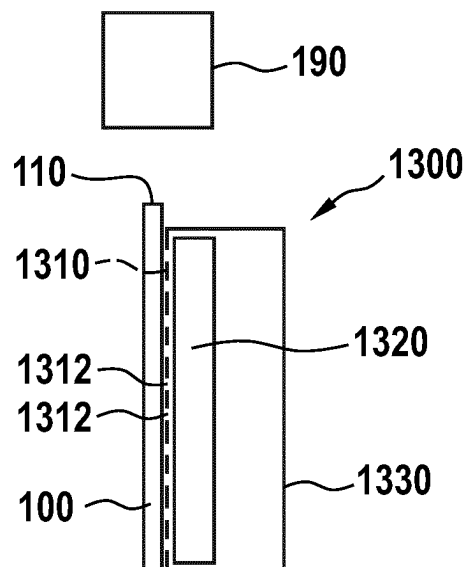
Fig. 1210**Fig. 13**

Fig. 14a

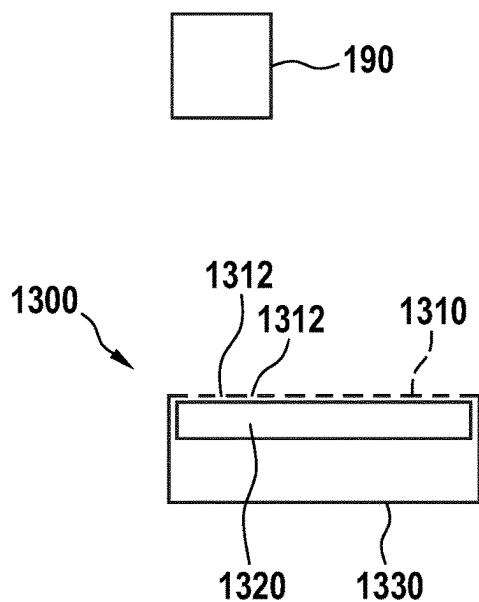


Fig. 14b

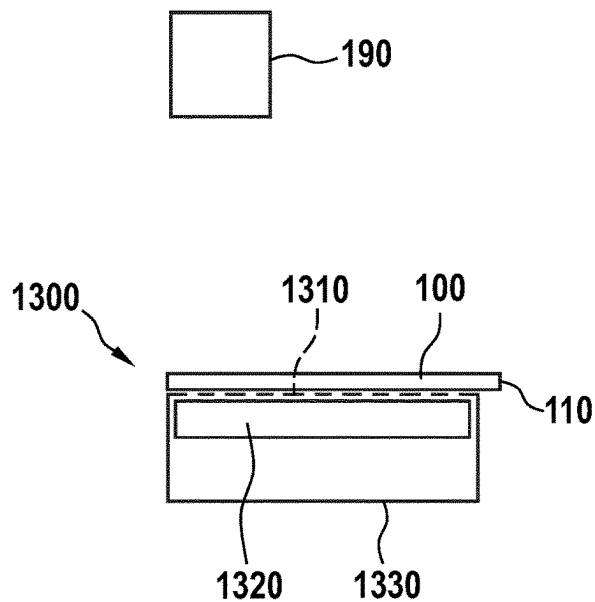


Fig. 14c

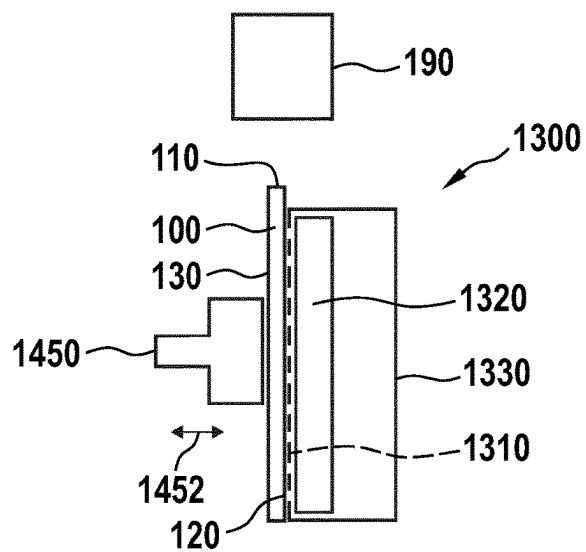


Fig. 15a

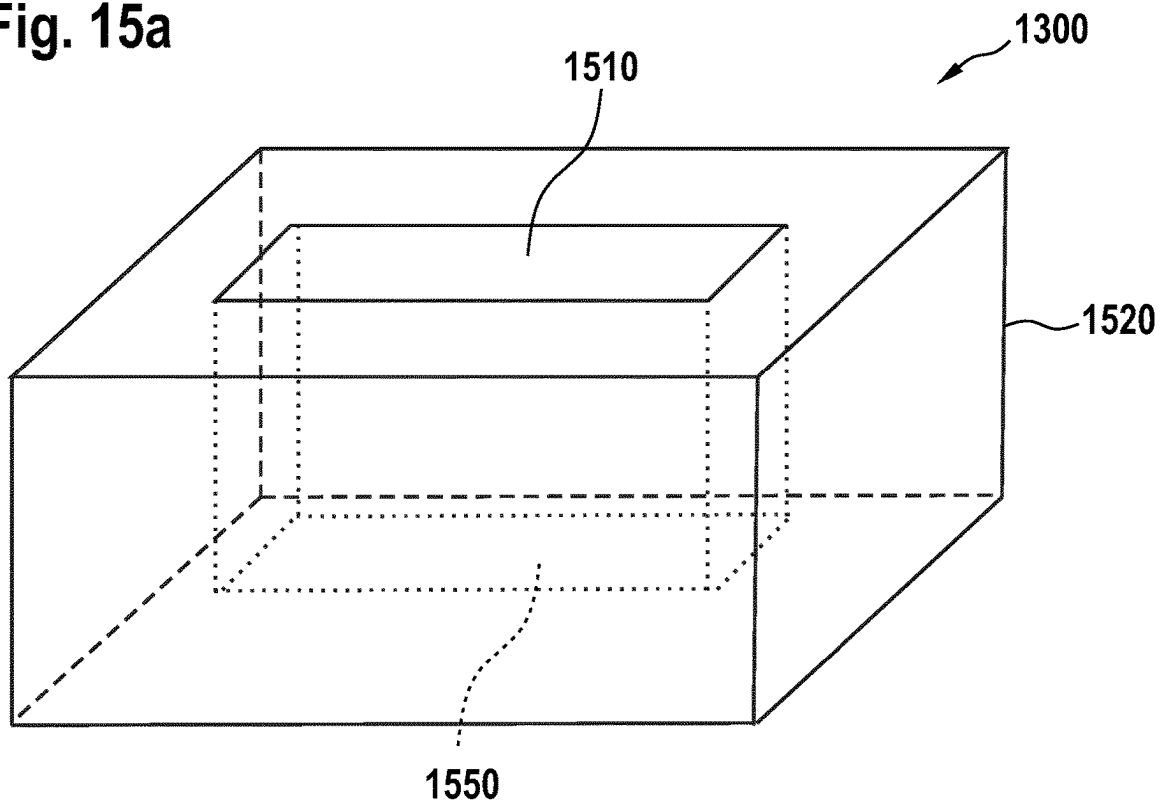


Fig. 15b

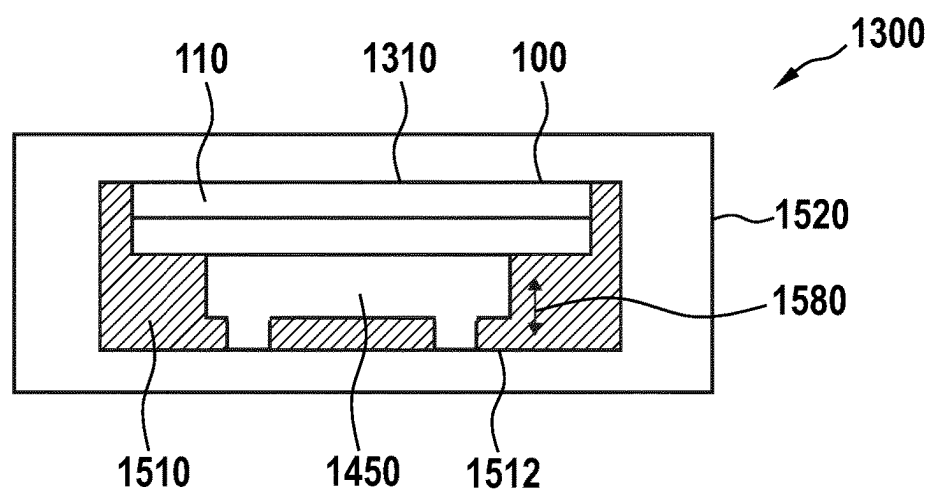


Fig. 16a

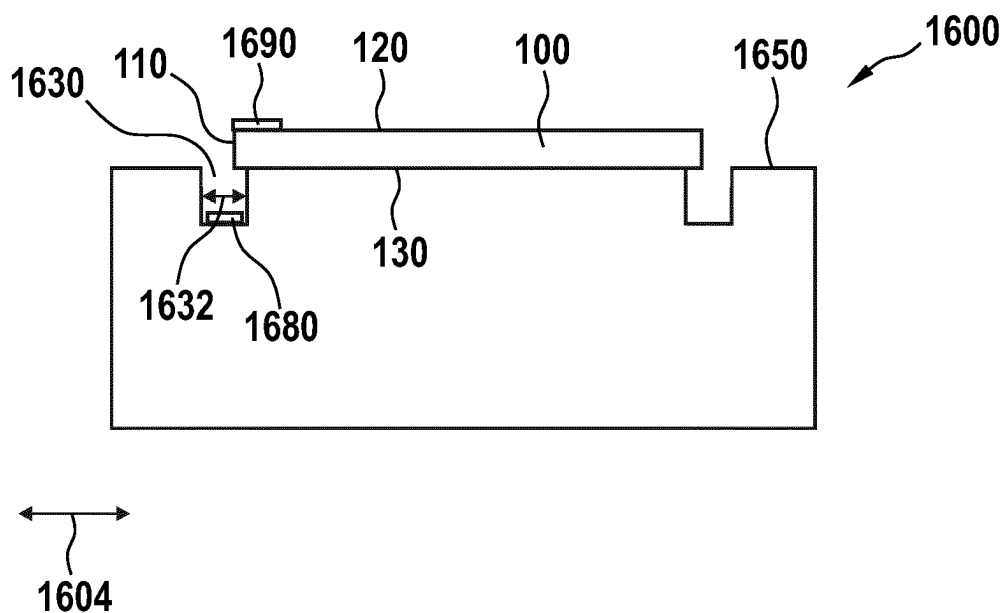


Fig. 16b

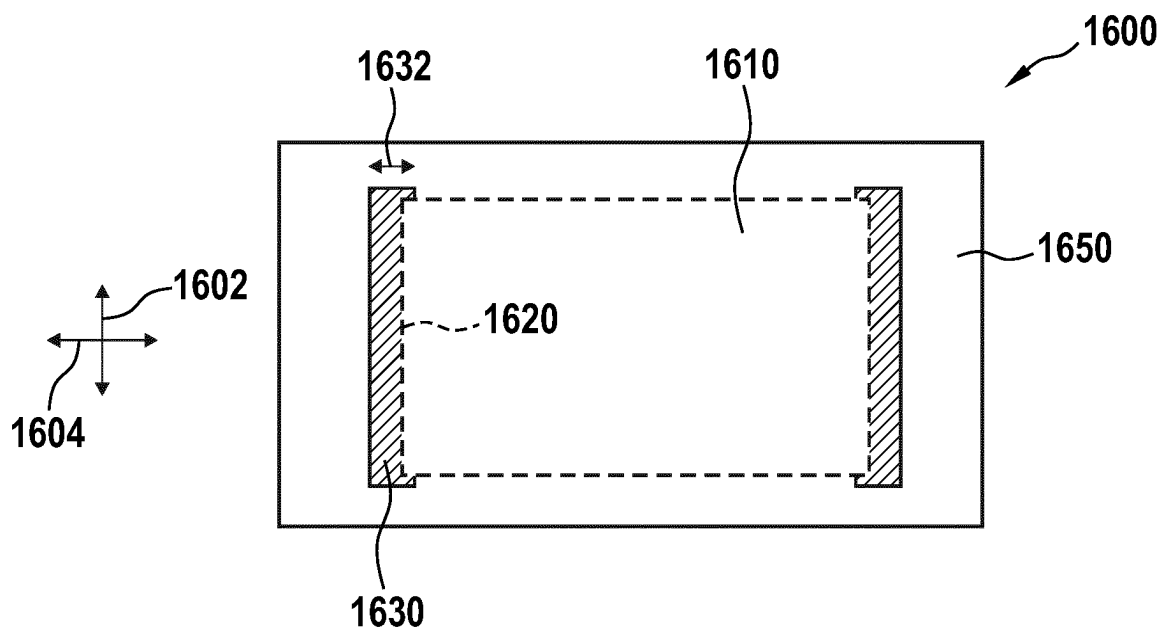


Fig. 17

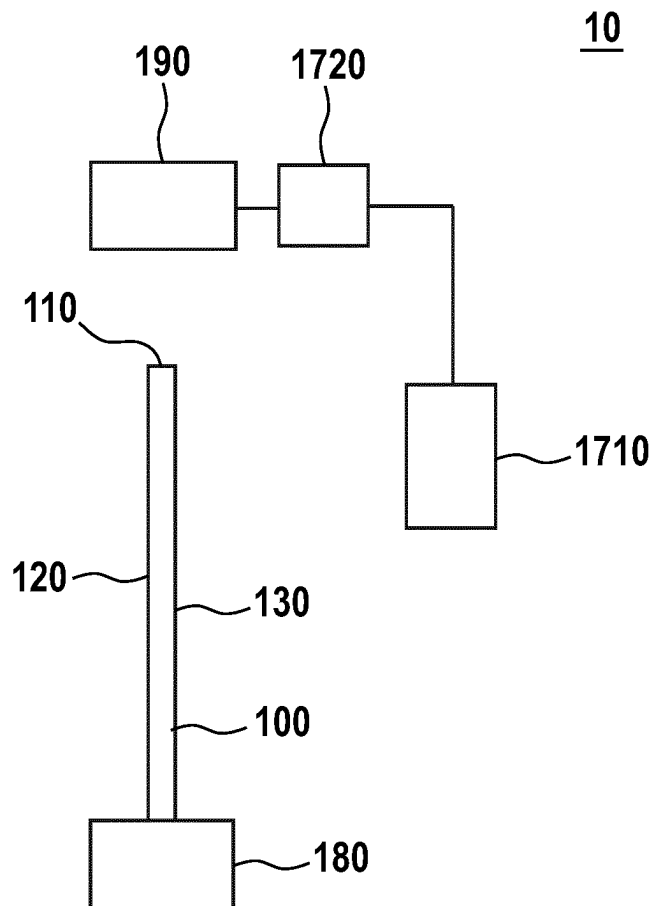


Fig. 18

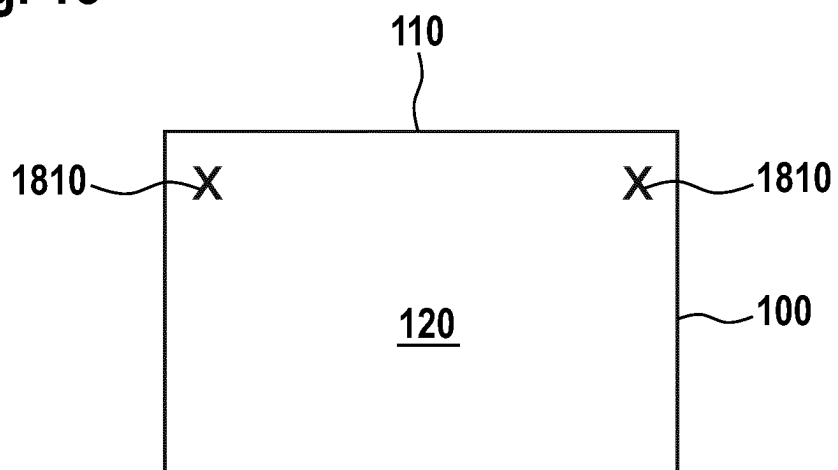


Fig. 19

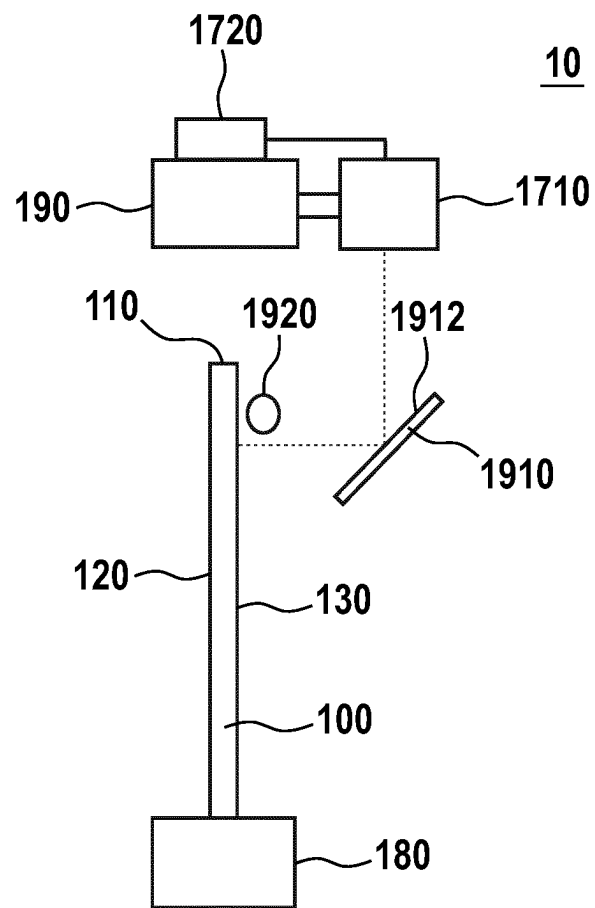


Fig. 20a

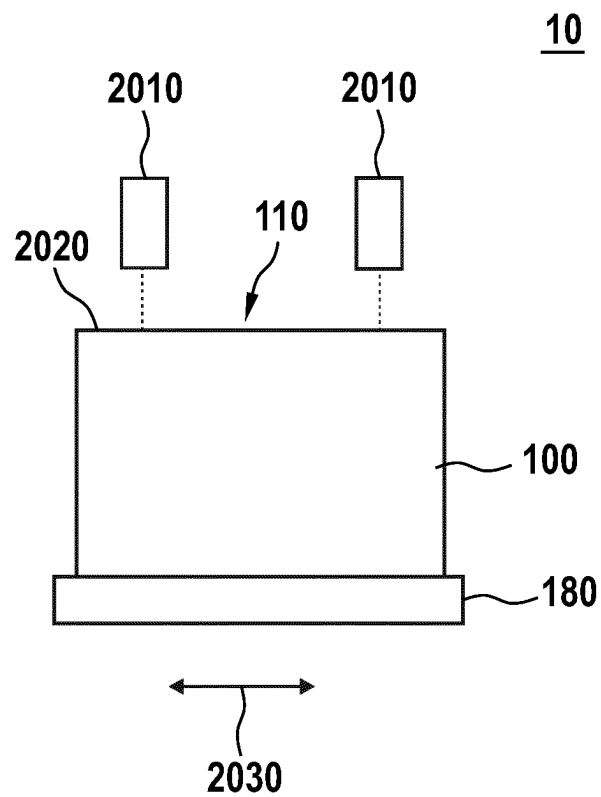
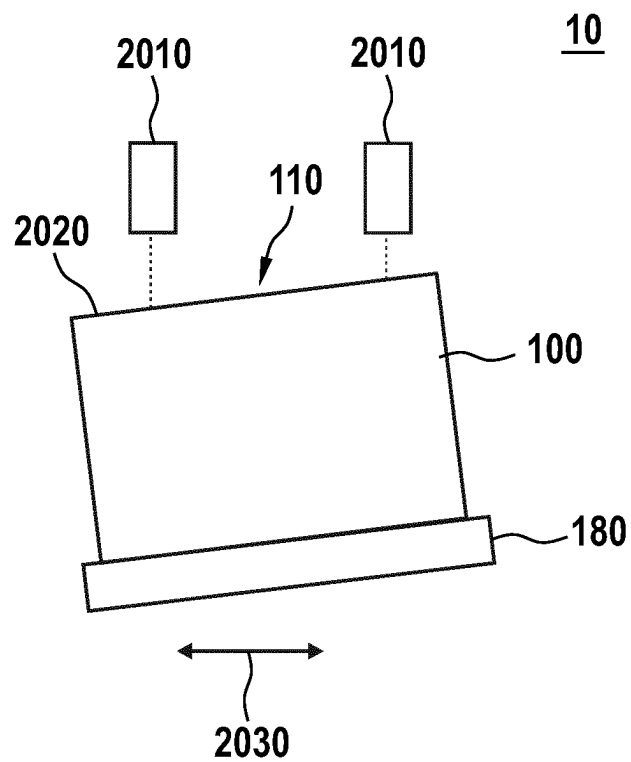


Fig. 20b



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DEPOSITION APPARATUS, METHOD OF DEPOSITION ON A SUBSTRATE, SUBSTRATE STRUCTURE AND SUBSTRATE SUPPORT

FIELD

Embodiments of the present disclosure relate to a deposition apparatus and a method for depositing a material on a substrate, more specifically for depositing a conductive pattern or a mask on a substrate. More specifically, embodiments of the present disclosure relate to a printing apparatus, such as a screen printing apparatus, for printing a material on a substrate.

BACKGROUND

Many technological applications involve the deposition of a material on various kinds of substrates. For example, a substrate may be a substrate used in display technology, a semiconductor substrate, or a printed circuit board substrate. A substrate may be an inflexible substrate such as glass substrate, or a flexible substrate. A substrate may have a variety of possible dimensions, ranging from smartwatch dimension to large area substrates for display manufacturing.

A layer of material can be deposited on a substrate, wherein the deposited layer forms a pattern of deposited material. Several deposition techniques can be used to deposit a pattern on a substrate, such as screen printing, physical vapor deposition or chemical vapor deposition. In some implementations, a patterned layer is formed by using a mask covering a portion of the substrate.

Several applications involve substrates on which conductive patterns are deposited. Such conductive patterns may include conductive materials deposited on different regions of the substrate. Conductive patterns may include a plurality of conductive portions having different shapes and sizes. Some conductive portions of a pattern may be connected to other conductive portions of the pattern by yet further conductive portions, which can result in conductive patterns having a complex structure.

In light of the above, there is a need for improving the deposition of patterned structures on substrates.

SUMMARY

According to an embodiment, a deposition apparatus is provided. The deposition apparatus includes a first substrate support for supporting a substrate in a substantially vertical orientation. The substrate has a first main surface, a second main surface opposite the first main surface and a side surface between the first main surface and the second main surface. The deposition apparatus includes a first deposition device for depositing a first conductive pattern or a first resist mask on the side surface of the substrate while the substrate is supported in the substantially vertical orientation by the first substrate support.

According to a further embodiment, a deposition apparatus is provided. The deposition apparatus includes a first printing device. The deposition apparatus includes a first orientation-changing device downstream of the first printing device. The deposition apparatus includes a second printing device downstream of the first orientation-changing device. The deposition apparatus includes a second orientation-changing device downstream of the second printing device.

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The deposition apparatus includes a third printing device downstream of the second orientation-changing device.

According to a further embodiment, a method of deposition on a substrate is provided. The method includes providing a substrate in a substantially vertical orientation. The substrate has a first main surface, a second main surface opposite the first main surface and a side surface between the first main surface and the second main surface. The method includes depositing a first conductive pattern or a first resist mask on the side surface of the substrate while the substrate is in the substantially vertical orientation.

According to a further embodiment, a substrate structure is provided. The substrate structure includes a substrate having a first surface, a second surface opposite the first surface and a side surface between the first surface and the second surface. The substrate structure includes a plurality of electrical connections connecting the first surface with the second surface. Each electrical connection includes a conductive line on the first surface, a conductive line on the side surface, and a conductive line on the second surface. The conductive line on the side surface contacts the conductive line on the first surface and the conductive line on the second surface.

According to a further embodiment, a substrate structure is provided. The substrate structure includes a substrate having a first surface, a second surface opposite the first surface and a side surface between the first surface and the second surface. The substrate surface includes a resist mask. The resist mask includes a first resist mask portion covering a portion of the side surface, the first resist mask portion having a first plurality of line-like openings. The resist mask includes a second resist mask portion covering a portion of the first surface, the second resist mask portion having a second plurality of line-like openings. The resist mask includes a third resist mask portion covering a portion of the second surface, the third resist mask portion having a third plurality of line-like openings. Each line-like opening of the first plurality of line-like openings is adjacent to a line-like opening of the second plurality of line-like openings and to a line-like opening of the third plurality of line-like openings.

According to a further embodiment, a substrate support for supporting a substrate in a substantially vertical orientation is provided. The substrate support includes a first surface, the first surface being a stationary surface which is substantially vertical or a movable surface which is movable between a first position and a second position, wherein in the second position the first surface is substantially vertical. The substrate support includes a substrate attraction device adjacent to the first surface.

According to a further embodiment, a substrate support for a deposition apparatus is provided. The substrate support includes a housing comprising a slit to receive a substrate in a substantially vertical orientation.

According to a further embodiment, a substrate support is provided. The substrate support includes a first surface, the first surface being a stationary surface which is substantially vertical or a movable surface which is movable between a first position and a second position, wherein in the second position the first surface is substantially vertical. The substrate support includes a biasing device for biasing a substrate against the first surface, the substrate having a first main surface, a second main surface opposite the first main surface and a side surface between the first main surface and the second main surface. The substrate support is configured for holding the substrate in a substantially vertical orientation.

tion in a manner such that the side surface of the substrate is exposed for deposition of a material on the side surface.

Embodiments are also directed at apparatuses for carrying out the disclosed methods and include apparatus parts for performing each described method aspect. These method aspects may be performed by way of hardware components, a computer programmed by appropriate software, by any combination of the two or in any other manner. Furthermore, embodiments according to the disclosure are also directed at methods for operating the described apparatus. The methods for operating the described apparatus include method aspects for carrying out every function of the apparatus.

BRIEF DESCRIPTION OF THE DRAWINGS

So that the manner in which the above recited features of the present disclosure can be understood in detail, a more particular description of the disclosure, briefly summarized above, may be had by reference to embodiments. The accompanying drawings relate to embodiments of the disclosure and are described in the following:

FIG. 1 shows a side view of a deposition apparatus according to embodiments described herein;

FIG. 2a shows an example of a substrate having a first conductive pattern on the side surface of the substrate;

FIG. 2b shows an example of a substrate having a first resist mask on the side surface of the substrate;

FIGS. 3a-c show a deposition apparatus according to embodiments described herein, the deposition apparatus including a first deposition device, a second deposition device and a third deposition device;

FIG. 4a shows an example of a substrate having a second conductive pattern on the first main surface of the substrate;

FIG. 4b shows an example of a substrate having a second resist mask on the first main surface of the substrate;

FIG. 5a shows an example of a substrate having a third conductive pattern on the second main surface of the substrate;

FIG. 5b shows an example of a substrate having a third resist mask on the second main surface of the substrate;

FIGS. 6-7 show examples of a substrate having a first, second and third conductive pattern;

FIG. 8 shows a substrate structure according to embodiments described herein;

FIG. 9 shows a deposition apparatus according to embodiments described herein;

FIG. 10 shows a deposition apparatus according to embodiments described herein;

FIGS. 11a-b show a top view of a stack of substrates;

FIG. 12 shows a deposition apparatus according to embodiments described herein;

FIG. 13 shows a substrate support according to embodiments described herein;

FIGS. 14a-c show a substrate support according to embodiments described herein;

FIGS. 15a-b show a substrate support according to embodiments described herein;

FIGS. 16a-b show a substrate support according to embodiments described herein;

FIG. 17 shows a deposition apparatus according to embodiments described herein;

FIG. 18 shows a substrate including alignment marks;

FIG. 19 shows a deposition apparatus according to embodiments described herein; and

FIGS. 20a-b shows a deposition apparatus according to embodiments described herein.

DETAILED DESCRIPTION

Reference will now be made in detail to the various embodiments of the disclosure, one or more examples of which are illustrated in the figures. Within the following description of the drawings, the same reference numbers refer to same components. Generally, only the differences with respect to individual embodiments are described. Each example is provided by way of explanation of the disclosure and is not meant as a limitation of the disclosure. Further, features illustrated or described as part of one embodiment can be used on or in conjunction with other embodiments to yield yet a further embodiment. It is intended that the description includes such modifications and variations.

FIG. 1 shows a side view of a deposition apparatus 10 according to embodiments described herein.

According to an embodiment, a deposition apparatus 10 is provided. The deposition apparatus 10 includes a first substrate support 180. The first substrate support 180 may be for supporting a substrate 100 in a substantially vertical orientation. The vertical direction is indicated in FIG. 1 by the arrow 1. The substrate 100 has a first main surface 120, a second main surface 130 opposite the first main surface 120 and a side surface 110 between the first main surface 120 and the second main surface 130. The deposition apparatus 10 includes a first deposition device 190. The first deposition device 190 may be for depositing a first conductive pattern 250 or a first resist mask 260 on the side surface 110 of the substrate 100 while the substrate 100 is supported in the substantially vertical orientation by the first substrate support 180.

FIGS. 2a-b show a top view of the side surface 110 of an exemplary substrate 100 as described herein. A first conductive pattern 250 is deposited on the side surface 110 of the substrate 100 shown in FIG. 2a. As shown, the first conductive pattern 250, indicated by the hashed portions in FIG. 2a, can include a plurality of conductive lines 252. A first resist mask 260 as described herein is deposited on the side surface 110 of the substrate 100 shown in FIG. 2b. As shown, the first resist mask 260, indicated by the hashed portions in FIG. 2b, can include a plurality of line-like openings 262. If the side surface 110 including the first resist mask 260 is coated with a conductive material, e.g. by sputtering or plating, conductive lines similar to the conductive lines 252 are formed in the line-like openings 262 of the first resist mask 260.

A substrate as described herein, such as e.g. the substrate 100, may be provided in a substantially vertical orientation. A substantially vertical orientation of the substrate includes small deviations from an exact vertical orientation, e.g. deviations of 5% or less or 10% or less, possibly even up to 15%.

A substrate as described herein may be a substantially planar or sheet-like substrate. For example, a substrate may be a substrate used in display technology, a semiconductor substrate, a printed circuit board substrate, or any kind of circuit for which it is beneficial to provide an electrical connection between two surfaces of the substrate.

The substrate may, e.g., be an inflexible substrate such as a glass substrate, or a flexible substrate. The substrate may have a variety of possible dimensions, ranging from smart-watch dimension to large area substrates for display manufacturing.

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A substrate as described herein may have a first main surface. The substrate may have a second main surface opposite the first main surface. The terms “first main surface” and the “second main surface” can be understood in a manner such that a total area of the substrate is substantially the same as the area of at least one of the first main surface and the second surface.

The substrate may have a side surface between the first main surface and the second main surface. The side surface may connect the first main surface with the second main surface. The area of the side surface is much smaller than the area of the first main surface and the area of the second main surface. The side surface may be a thin strip-like surface between the first major surface and the second major surface.

The substrate may have a thickness, e.g. a thickness of 20 mm or less, particularly 5 mm or less, more particularly 1 mm or less, such as e.g. about 0.5 mm. The thickness may be as small as e.g. 0.1 mm. The distance between the first main surface and the second main surface may correspond to the thickness of the substrate. The side surface may have a length and a width. The length is larger than the width. The width of the side surface may correspond to the thickness of the substrate. For example, the width of the side surface may be 5 mm or less, more particularly 1 mm or less, such as e.g. about 0.5 mm.

The first deposition device **190**, or at least a portion thereof, may be arranged above the first substrate support **180**. While the substrate **100** is supported by the first substrate support **180**, the side surface **110** may be located at a top portion of the substrate **100**. The first deposition device **190** may be arranged to deposit the first conductive pattern **250** onto the side surface **110** from above.

A conductive pattern as described herein, such as e.g. the first conductive pattern **250**, the second conductive line pattern **450** or the third conductive line pattern **550**, may include a plurality of conductive lines. The plurality of conductive lines may include, e.g., 10 or more lines per mm. The plurality of conductive lines may include a first conductive line. The plurality of conductive lines may include a second conductive line.

The first conductive line may be substantially parallel to the second conductive line. The term “substantially parallel” includes conductive lines having an angle from 0 degrees to 15 degrees with respect to each other. The plurality of conductive lines may be a plurality of substantially parallel lines. In other implementations, the conductive lines may not be substantially parallel. A conductive pattern can include conductive lines having an angle relative to each other, e.g., an angle of 20 degrees or more.

According to embodiments described herein, a conductive pattern may be printed on the substrate. Printing results in thicker conductive patterns as compared to other deposition techniques such as e.g. sputtering. Thicker conductive patterns provide the advantage that a more reliable, i.e. electrically and mechanically more stable electrical connection can be formed. In order to further increase thickness, multiple passes can be applied. A conductive pattern as described herein may have a thickness of 20 μm or less and/or 0.5 μm or more. For example, the thickness can be 3 μm or more.

Further, printing provides the advantage that the printed patterns have a good adhesion to the substrate.

The first conductive pattern **250** as described herein may include a conductive line **252**. The conductive line **252** may extend from a first edge region of the side surface **110** to a second edge region of the side surface **110**. The first edge

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region may be adjacent to the first main surface **120**. The first edge region may be a region where the side surface **110** meets the first main surface **120**. The second edge region may be opposite the first edge region. The second edge region may be adjacent to the second main surface **130**. The second edge region may be a region where the side surface **110** meets the second main surface **130**.

In some implementations, the conductive line **252** may be substantially perpendicular to at least one of the first main surface **120** and the second main surface **130**. The term “substantially perpendicular” includes an angle of 90 degrees and small deviations therefrom, e.g. angles from 75 degrees to 105 degrees. In other implementations, the conductive line **252** may be at an angle with respect to at least one of the first main surface and the second main surface, e.g. an angle of 70 degrees or less. The first conductive pattern **250** as described herein may include a plurality of conductive lines. Each conductive line **252** of the plurality of conductive lines may extend from the first edge region of the side surface **110** to the second edge region of the side surface **110**. Each conductive line **252** may, for example, be substantially perpendicular to at least one of the first main surface **120** and the second main surface **130**.

A conductive line as described herein, e.g. a conductive line of the first conductive pattern **250**, the second conductive pattern **450** or the third conductive pattern **550**, may have a line width of 200 μm or less and/or 5 μm or more. For example, the line width may be from 20 to 80 μm . A conductive line may have a line length of 500 μm or less and/or 50 μm or more, such as e.g. a line length of about 100 μm .

A resist mask, such as the first resist mask **260**, the second resist mask **460** or the third resist mask **560** as described herein, may cover one or more portion(s) of the substrate. One or more other portions of the substrate may not be covered by the resist mask. The resist mask may be shaped in the form of a pattern. The pattern may be a “negative image” of a target pattern of conductive material that is to be deposited on the substrate. For example, the resist mask may have a plurality of openings shaped in the form of a plurality of lines. The plurality of lines may correspond to a target pattern that is to be deposited on the substrate. After the resist mask is deposited on the substrate, the substrate including the resist mask may be coated, e.g. in a sputtering or plating process, to deposit a conductive material on the substrate. The conductive material is deposited on the resist mask and on the portion(s) of the substrate not covered by the resist mask. No conductive material is deposited on the portion(s) of the substrate covered by the resist mask. In light thereof, the conductive material may be deposited on the substrate in the form of a target pattern determined by the shape of the resist mask. After the deposition process, the resist mask may be removed from the substrate, e.g. in a chemical removal process. A substrate having a target pattern of conductive material formed thereon is provided.

A resist mask as described herein may include a plurality of line-like openings, e.g. openings having the shape of a thin strip or slits, in the resist mask. The plurality of line-like openings of the resist mask may correspond to a plurality of lines of a conductive pattern to be deposited on the substrate using the resist mask. The plurality of line-like openings may include, e.g. 10 or more line-like openings per mm. The plurality of line-like openings may include a first line-like opening or slit. The plurality of line-like openings may include a second line-like opening or slit. The first line-like opening may be substantially parallel to the second line-like opening. The plurality of line-like openings may be a

plurality of substantially parallel line-like openings. Alternatively, the line-like openings may be at an angle relative to each other, e.g. an angle of 20 degrees or more.

The first resist mask 260 as described herein may include a line-like opening 262 or slit. The line-like opening 262 may extend from a first edge region, as described herein, of the side surface 110 to a second edge region, as described herein, of the side surface 110. The line-like opening 262 may be substantially perpendicular to at least one of the first main surface 120 and the second main surface 130. Alternatively, the line-like opening 262 may be at an angle with respect to at least one of the first main surface 120 and the second main surface 130, e.g. an angle of 70 degrees or less.

The first resist mask 260 as described herein may include a plurality of line-like openings. Each line-like opening 262 of the plurality of line-like openings may extend from the first edge region of the side surface 110 to the second edge region of the side surface 110. Each line-like opening 262 may be substantially perpendicular to at least one of the first main surface and the second main surface.

The first deposition device 190 may include or be a printing device. The first deposition device 190 may be configured to print the first conductive pattern 250 and/or the first resist mask 260 on the side surface 110 of the substrate. The first deposition device may be a screen printing device. The first deposition device may include a screen for screen printing. The screen may be arranged above the first substrate support 180. The first deposition device 190 may be configured to print the first conductive pattern 250 and/or the first resist mask 260 on the side surface 110 of the substrate 100 by urging a printing material through the screen. For printing the first conductive pattern 250 (or the second conductive pattern 450 or the third conductive pattern 550, or any other conductive pattern described herein), the printing material urged through the screen may be a conductive material, e.g. a conductive ink or paste, such as a silver or copper ink, or any other kind of conductive screen printing material. For printing the first resist mask 260 (or the second resist mask 460 or the third resist mask 560 as described herein), the printing material urged through the screen may be an insulating material, e.g. a polymer.

A deposition apparatus 10 according to embodiments described herein may include a layer deposition device for depositing a conductive material on the substrate. The layer deposition device may be configured to deposit the conductive material on at least the side surface 110 of the substrate 100 including the first resist mask 260. The layer deposition device may be, e.g., a sputtering device or a plating device.

FIGS. 3a-c show a deposition apparatus 10 according to embodiments described herein. The deposition apparatus 10 includes a first deposition device 190, a second deposition device 330 and a third deposition device 370. The deposition apparatus 10 includes a first substrate support 180, a second substrate support 320 and a third substrate support 360. As shown in FIG. 3a, the substrate 100 can be supported by the second substrate support 320 in a substantially horizontal orientation. The first main surface 120 of the substrate 100 can face upward. The second deposition device 330 can deposit a second conductive pattern or a second resist mask on the first main surface 120 of the substrate 100. The substrate 100 can be moved from the second substrate support 320 to the third substrate support 360. As shown in FIG. 3b, the substrate 100 can be supported by the third substrate support 360 in a substantially horizontal orientation. The second main surface 130 of the substrate 100 can face upward. The third deposition device 370 can deposit a third conductive pattern or a third resist mask on the second

main surface 130 of the substrate 100. The substrate 100 can be moved from the third substrate support 360 to the first substrate support 180. The first deposition device 190 can deposit the first conductive pattern 250 on the side surface 110 of the substrate 100.

FIGS. 4a-b show a top view of the first main surface 120 of an exemplary substrate 100. A second conductive pattern 450 as described herein is deposited on the first main surface 120 shown in FIG. 4a. As shown, the second conductive pattern 450, indicated by the hashed portions in FIG. 4a, can include a plurality of conductive lines 452. A second resist mask 460 as described herein is deposited on the first main surface 120 shown in FIG. 4b. As shown, the second resist mask 460, indicated by the hashed portions in FIG. 4b, can include a plurality of line-like openings 462.

FIGS. 5a-b show a top view of the second main surface 130 of an exemplary substrate 100. A third conductive pattern 550 including a plurality of conductive lines 552 is deposited on the second main surface 130 shown in FIG. 5a. A third resist mask 560 including a plurality of line-like openings 562 is deposited on the second main surface 130 shown in FIG. 5b.

A deposition apparatus 10 according to embodiments described herein may include a second deposition device 330 for depositing a second conductive pattern 450 or a second resist mask 460 on the first main surface 120 of the substrate 100. A portion of the second conductive pattern 450 may contact a portion of the first conductive pattern 250. A portion of the second resist mask 460 may contact a portion of the first resist mask 260.

The second deposition device 330 may include or be a printing device, particularly a screen printing device. The second deposition device 330 may be configured to print the second conductive pattern 450 and/or the second resist mask 460 on the first main surface 120 of the substrate 100.

A deposition apparatus 10 according to embodiments described herein may include a second substrate support 320. The second substrate support 320 may be configured for supporting the substrate 100 in a substantially horizontal orientation. The second deposition device 330 may be arranged for depositing the second conductive pattern 450 or the second resist mask 460 on the first main surface 120 of the substrate 100 while the substrate 100 is supported by the second substrate support 320.

The second conductive pattern 450 as described herein may include a conductive line 452. The conductive line 452 may extend from a first end of the conductive line 452 to a second end of the conductive line 452. The first end of the conductive line 452 may be located at, or adjacent to, a region where the first main surface 120 meets the side surface 110. The conductive line 452 of the second conductive pattern 450 may contact a conductive line 252 of the first conductive pattern 250. An electric current can flow from the conductive line 452 to the conductive line 252 or vice versa. The conductive line 452 together with the conductive line 252 may form an electrical connection from the first main surface 120 to the side surface 110. The second conductive pattern 450 may include a plurality of conductive lines. The plurality of conductive lines may be substantially parallel conductive lines or conductive lines at an angle relative to each other. Each conductive line 452 of the plurality of conductive lines may extend from a first end of the conductive line 452 to a second end of the conductive line 452. The first end of each conductive line 452 may be located at, or adjacent to, a region where the first main surface 120 meets the side surface 110. Each conductive line 452 of the plurality of conductive lines may contact a

corresponding conductive line 252 of the first conductive pattern 250. A plurality of electrical connections between the side surface 110 and the first main surface 120 may be formed.

The second resist mask 460 as described herein may include a line-like opening 462 or slit. The line-like opening 462 may extend from a first end of the line-like opening 462 to a second end of the line-like opening 462. The first end of the line-like opening 462 may be located at, or adjacent to, a region where the first main surface 120 meets the side surface 110. The first end of the line-like opening 462 of the second resist mask 460 may be adjacent to an end of a line-like opening 262 of the first resist mask 260. After the side surface 110 including the first resist mask 260 and the first main surface 120 including the second resist mask 460 are coated with a conductive material, a conductive line is formed in the line-like opening 262 of the first resist mask 260 and a conductive line is formed in the line-like opening 462 of the second resist mask 460, wherein the two conductive lines contact each other so that a current can flow between the two conductive lines. The second resist mask 460 may include a plurality of line-like openings. The plurality of line-like openings may be substantially parallel to each other or may be at an angle relative to each other. Each line-like opening 462 of the plurality of line-like openings may extend from a first end of the line-like opening 462 to a second end of the line-like opening 462. The first end of each line-like opening 462 may be located at, or adjacent to, a region where the first main surface 120 meets the side surface 110. The first end of each line-like opening 462 of the second resist mask 460 may be adjacent to, or in communication with, an end of a corresponding line-like opening 262 of the first resist mask 260.

A deposition apparatus 10 according to embodiments described herein may include a third deposition device 370 for depositing a third conductive pattern 550 or a third resist mask 560 on the second main surface 130 of the substrate 100. A portion of the third conductive pattern 550 may contact a portion of the first conductive pattern 250. A portion of the third resist mask 560 may contact a portion of the first resist mask 260.

The third deposition device 370 may include or be a printing device, particularly a screen printing device. The third deposition device 370 may be configured to print the third conductive pattern 550 and/or the third resist mask 560 on the second main surface 130 of the substrate 100.

A deposition apparatus 10 according to embodiments described herein may include a third substrate support 360. The third substrate support 360 may be configured for supporting the substrate 100 in a substantially horizontal orientation. The third deposition device 370 may be arranged for depositing the third conductive pattern 550 or the third resist mask 560 on the second main surface 130 of the substrate 100 while the substrate 100 is supported by the third substrate support 360.

The third conductive pattern 550 as described herein may include a conductive line 552. The conductive line 552 may extend from a first end of the conductive line 552 to a second end of the conductive line 552. The first end of the conductive line 552 may be located at, or adjacent to, a region where the second main surface 130 meets the side surface 110.

The conductive line 552 of the third conductive pattern 550 may contact a conductive line 252 of the first conductive pattern 250. The conductive line 252 of the first conductive pattern 250 may contact a conductive line 452 of the second conductive pattern 450, as described herein. The conductive

line 452 together with the conductive line 252 and the conductive line 552 may form an electrical connection from the first main surface 120 to the second main surface 130 via the side surface 110. In light thereof, embodiments described herein provide the advantage that an electrical connection can be formed between the first main surface 120 and the second main surface 130 without the need for a through-hole or via in the substrate 100 to establish the electrical connection.

The first conductive pattern 250 as described herein may include a first plurality of conductive lines. The second conductive pattern 450 may include a second plurality of conductive lines. The third conductive pattern 550 may include a third plurality of conductive lines. Each conductive line 252 of the first plurality of conductive lines may contact a conductive line 452 of the second plurality of conductive lines and a conductive line 552 of the third plurality of conductive lines to form an electrical connection from the first main surface 120 to the second main surface 130. In light thereof, embodiments described herein provide the advantage that a plurality of electrical connections can be formed between the first main surface 120 and the second main surface 130 without the need for through-holes or via in the substrate 100 to establish the electrical connections.

The third resist mask 560 as described herein may include a line-like opening 562 or slit. The line-like opening 562 may extend from a first end of the line-like opening 562 to a second end of the line-like opening 562. The first end of the line-like opening 562 may be located at, or adjacent to, a region where the second main surface 130 meets the side surface 110. The first end of the line-like opening 562 of the third resist mask 560 may be adjacent to an end of a line-like opening 262 of the first resist mask 260. The line-like opening 262 of the first resist mask 260 may be adjacent to a line-like opening 462 of the second resist mask 460, as described herein. After the side surface 110 including the first resist mask 260, the first main surface 120 including the second resist mask 460 and the second main surface 130 including the third resist mask 560 are coated with a conductive material, a conductive line is formed in the line-like opening 262 of the first resist mask 260, a conductive line is formed in the line-like opening 462 of the second resist mask 460 and a conductive line is formed in the line-like opening 562 of the third resist mask 560. The three conductive lines form an electrical connection from the first main surface 120 to the second main surface 130 via the side surface 110.

The first resist mask 260 as described herein may include a first plurality of line-like openings. The second resist mask 460 may include a second plurality of line-like openings. The third resist mask 560 may include a third plurality of line-like openings. Each line-like opening 262 of the first plurality of line-like openings may be adjacent to, or in communication with, a line-like opening 462 of the second plurality of line-like openings and a line-like opening 562 of the third plurality of line-like openings. When a conductive material is deposited in the line-like opening 262 and in the adjacent line-like openings 462 and 562, an electrical connection from the first main surface 120 to the second main surface 130 via the side surface 110 is formed. In light thereof, by coating the side surface 110 including the first resist mask 260, the first main surface 120 including the second resist mask 460 and the second main surface 130 including the third resist mask 560 with a conductive material, a plurality of electrical connections from the first main surface 120 to the second main surface 130 are formed.

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FIGS. 6-7 show a substrate structure 600 according to embodiments described herein. FIG. 6 shows an example of a substrate structure 600 resulting from depositing the first conductive pattern 250 on the side surface 110 after depositing the second conductive pattern 450 on the first main surface 120 and the third conductive pattern 550 on the second main surface 130. FIG. 7 shows an example of a substrate structure 600 resulting from depositing the first conductive pattern 250 before depositing the second conductive pattern 450 and the third conductive pattern 550. As shown in FIGS. 6-7, a conductive line 252 of the first conductive pattern 250 is deposited on the side surface 110. A conductive line 452 of the second conductive pattern 450 is deposited on the first main surface 120. A conductive line 552 of the third conductive pattern 550 is deposited on the second main surface 130. The conductive line 252 contacts the conductive line 452 and the conductive line 552. The conductive line 252 together with the conductive line 452 and the conductive line 552 provide an electrical connection from the first main surface 120 to the second main surface 130 via the side surface 110. In light thereof, embodiments described herein provide the advantage that an electrical connection from the first main surface 120 to the second main surface 130 can be provided while avoiding the formation of a through-hole through the substrate 100.

FIG. 8 shows a substrate structure 600 according to embodiments described herein. The substrate structure 600 includes a plurality of electrical connections between the first main surface 120 and the second main surface 130. Each electrical connection includes a conductive line 452 on the first main surface 120, a conductive line 252 on the side surface 110, and a conductive line 552 on the second main surface 130. The conductive line 252 on the side surface 110 contacts the conductive line 452 on the first main surface 120 and the conductive line 552 on the second main surface 130.

FIG. 9 shows a deposition apparatus 10 according to embodiments described herein. In FIG. 9, substrates are processed from left to right. The deposition apparatus 10 includes a storing unit 910, e.g. a cassette, for storing a plurality of substrates. The deposition apparatus 10 includes the second deposition device 330 as described herein, e.g. for depositing the second conductive pattern 450 or the second resist mask 460 on the first main surface 120 of the substrate 100. The deposition apparatus 10 includes an inspection device 912, e.g. an optical inspection device, for inspecting the second conductive pattern 450 or the second resist mask 460. The deposition apparatus 10 includes a drying device 914 for drying the second conductive pattern 450 or the second resist mask 460. The deposition apparatus 10 includes an orientation-changing device 916, e.g. for flipping the substrate 100 by about 180 degrees from a first horizontal orientation to a second horizontal orientation.

The deposition apparatus 10 shown in FIG. 9 includes the third deposition device 370 as described herein, e.g. for depositing the third conductive pattern 550 or the third resist mask 560 on the second main surface 130 of the substrate 100. The deposition apparatus 10 includes an inspection device 922, e.g. an optical inspection device, for inspecting the third conductive pattern 550 or the third resist mask 560. The deposition apparatus 10 includes a drying device 924 for drying the third conductive pattern 550 or the third resist mask 560. The deposition apparatus 10 includes an orientation-changing device 926, e.g. for flipping the substrate 100 by about 90 degrees from a horizontal orientation to a vertical orientation.

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The deposition apparatus 10 shown in FIG. 9 includes the first deposition device 190 as described herein, e.g. for depositing the first conductive pattern 250 or the first resist mask 260 on the side surface 110 of the substrate 100. The deposition apparatus 10 includes an inspection device 932, e.g. an optical inspection device, for inspecting the first conductive pattern 250 or the first resist mask 260. The deposition apparatus 10 includes a drying device 934 for drying the first conductive pattern 250 or the first resist mask 260.

The deposition apparatus 10 shown in FIG. 9 is an exemplary embodiment shown for the purpose of illustration. Several of the components shown in FIG. 9 can be omitted. For example, the storing unit 910, the inspection devices 912, 922 and 932, and the drying devices 914, 924 and 934 are optional components and can be omitted. Further, the deposition apparatus 10 can include additional components not shown in FIG. 9, such as an unloading unit for unloading the substrates. Further, the order of processing the surfaces of the substrates is also exemplary and can be changed, e.g. by performing printing on the side surface first and then on the two main surfaces, or any other order depending on the application.

A deposition apparatus 10 as described herein may include a first orientation-changing device, e.g. the orientation-changing device 926 shown in FIG. 9, for changing the orientation of the substrate 100 from a first orientation of the substrate to a second orientation of the substrate. The first orientation-changing device may be configured for changing the orientation of the substrate 100 by an angle of about 90 degrees. The first orientation-changing device may be configured for changing the orientation of the substrate 100 from a substantially vertical orientation to a substantially horizontal orientation and/or from a substantially horizontal orientation to a substantially vertical orientation. The term "substantially horizontal" includes a substrate orientation which can deviate from exact horizontality by an angle of 5 degrees or less, 10 degrees or less, or in some cases even up to 15 degrees.

A deposition apparatus 10 as described herein may include a second orientation-changing device, e.g. orientation-changing device 916 shown in FIG. 9, for changing the orientation of the substrate 100. The second orientation-changing device may be configured for changing the orientation of the substrate 100 by an angle of about 180 degrees. The second orientation-changing device may be configured for changing the orientation of the substrate 100 from a first substantially horizontal orientation to a second substantially horizontal orientation. In the first substantially horizontal orientation, the first main surface 120 of the substrate 100 may face upward. In the second substantially horizontal orientation, the second main surface 130 of the substrate 100 may face upward.

The first orientation-changing device may be arranged upstream of the first deposition device 190. The second orientation-changing device may be arranged upstream of the first orientation-changing device.

The terms "upstream" and "downstream" are defined relative to a processing direction, or processing flow, of the deposition apparatus 10.

In some implementations, the two main surfaces of the substrate are provided with a conductive pattern first, followed by patterning the side surface of the substrate. The substrate 100 may be processed by depositing a conductive pattern on the first main surface 120 in a first substantially horizontal orientation of the substrate 100, followed by flipping the substrate 100, followed by depositing a conduc-

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tive pattern on the second main surface **130** in a second substantially horizontal orientation of the substrate **100**, followed by changing the orientation of the substrate **100** by about 90 degrees, followed by depositing a conductive pattern on the side surface **110** in a substantially vertical orientation of the substrate **100**.

In other implementations, the surfaces of the substrate may be processed in a different order. For example, the substrate **100** may be processed by depositing a conductive pattern on the first main surface **120** in a first substantially horizontal orientation of the substrate **100**, followed by changing the orientation of the substrate **100** by about 90 degrees, followed by depositing a conductive pattern on the side surface **110** in a substantially vertical orientation of the substrate **100**, followed by changing the orientation of the substrate **100** by another 90 degrees, followed by depositing a conductive pattern on the second main surface **130** in a second substantially horizontal orientation of the substrate **100**. Any other order of processing the surfaces of the substrate is also possible. The same applies to the order for depositing the resist masks on the surfaces of the substrate.

FIG. **10** shows a deposition apparatus **10** according to embodiments described herein.

The first substrate support **180** as described herein may be configured for supporting a stack of substrates **1010** in a substantially vertical orientation. The stack of substrates **1010** in the substantially vertical orientation may have a top portion **1012**. The first deposition device **190** may be configured for depositing, particularly printing, a conductive pattern **1150** or a resist mask **1160** on the top portion **1012** of the stack of substrates **1010** while the stack of substrates **1010** is supported in the substantially vertical orientation by the first substrate support **180**.

FIG. **11a**, showing a top view of a stack of substrates **1010**, shows an example of a conductive pattern **1150** (hashed) on the top portion **1012** of the stack of substrates **1010**. FIG. **11b** shows an example of a resist mask **1160** (hashed) on the top portion **1012** of the stack of substrates **1010**.

The stack of substrates **1010** as described herein may include two, three, four, five or more substrates. The stack of substrates **1010** may include the substrate **100** as described herein. The stack of substrates can be held together, e.g., by a clamping device.

The stack of substrates **1010** can include a plurality of separators. Each separator can be arranged between two adjacent substrates of the stack of substrates **1010**. Each separator can be a sheet-like separator. For example, each separator can be a sheet of paper. Providing the separators between the substrates in the stack has several advantages. For example, the number of fully printed lines can be increased, a sliding of the substrates can be facilitated so that alignment of the substrates is easier, the separators can act as soft layers between the substrates so that particle generation is reduced, and the separators facilitate substrate detachment after drying or curing of the printed material.

Each substrate of the stack of substrates **1010** may have a first main surface, a second main surface and a side surface between the first main surface and the second main surface. While the stack of substrates **1010** is supported in the substantially vertical orientation by the first substrate support, the side surface of each substrate in the stack of substrates **1010** may be at the top portion of the stack of substrates **1010**. The top portion **1012** of the stack of substrates **1010** may include, or be formed by, the side surfaces of the substrates in the stack of substrates **1010**.

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The conductive pattern **1150** or resist mask **1160** may be deposited on the side surfaces of a plurality of substrates, or even all substrates, of the stack of substrates **1010**. The stack of substrates **1010** may include the substrate **100** as described herein. The stack of substrates **1010** may include a second substrate. A first portion of the conductive pattern **1150** or of the resist mask **1160** may be deposited on the side surface **110** of the substrate **100**. For example, the first portion of the conductive pattern **1150** may be the first conductive pattern **250** as described herein, or the first portion of the resist mask **1160** may be the first resist mask **260** as described herein. A second portion of the conductive pattern **1150** or of the resist mask **1160** may be deposited on a side surface of the second substrate. In light thereof, embodiments described herein provide the advantage that the side surfaces of multiple substrates can be provided with a conductive pattern or resist mask in a single processing cycle. The throughput of the deposition apparatus can be increased.

The conductive pattern **1150** may include a plurality of conductive lines **1152**. Each conductive line **1152** of the plurality of conductive lines may extend across multiple side surfaces of substrates of the stack of substrates **1010**. The resist mask **1160** may include a plurality of line-like openings **1162**. Each line-like opening **1162** of the plurality of line-like openings may extend across multiple side surfaces of substrates of the stack of substrates **1010**.

After depositing the conductive pattern **1150** or the resist mask **1160** on the stack of substrates **1010**, the substrates of the stack of substrates **1010** can be further processed one by one. For example, the substrate **100** can be taken as a single substrate from the stack of substrates **1010**. Therein, the substrate **100** includes the first conductive pattern **250** or the first resist mask **260** resulting from the deposition of the conductive pattern **1150** or the resist mask **1160** on the stack of substrates **1010**. After the substrate **100** is taken from the stack of substrates **1010**, the first main surface **120** of the substrate **100** can be provided with the second conductive pattern **450** or the second resist mask **460** as described herein. The second main surface **130** of the substrate **100** can be provided with the third conductive pattern **550** or the third resist mask **560** as described herein.

FIG. **12** shows a deposition apparatus **10** according to embodiments described herein.

According to a further embodiment, a deposition apparatus **10** is provided. The deposition apparatus includes a first printing device **1210**. The deposition apparatus **10** includes a first orientation-changing device **1220** downstream of the first printing device. The deposition apparatus **10** includes a second printing device **1230** downstream of the first orientation-changing device. The deposition apparatus **10** includes a second orientation-changing device **1240** downstream of the second printing device. The deposition apparatus **10** includes a third printing device **1250** downstream of the second orientation-changing device.

The terms “upstream” and “downstream” are defined relative to a processing direction, or processing flow, of the deposition apparatus **10**. For example, that the first orientation-changing device **1220** is downstream of the first printing device **1210** can be understood in the sense that the first orientation-changing device **1220** is arranged relative to the first printing device **1210** in a manner such that substrates are processed by the first printing device **1210** and thereafter transferred to the first orientation-changing device **1220** and processed by the first orientation-changing device **1220**.

As described herein, the first main surface **120**, the second main surface **130** and the side surface **110** of a substrate **100**

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may be processed in different orders. For example, in an implementation where the two main surfaces of the substrate are provided with a conductive pattern before the side surface is provided with a conductive pattern, the first printing device **1210** may be the second deposition device **330** as described herein, the second printing device **1230** may be the third deposition device **370** as described herein, and the third printing device **1250** may be the first deposition device **190** as described herein. In another implementation, the side surface **110** may be processed in between processing of the first main surface **120** and the second main surface **130**. In such an implementation, the first printing device **1210** may be the second deposition device **330** as described herein, the second printing device **1230** may be the first deposition device **190** as described herein, and the third printing device **1250** may be the third deposition device **370** as described herein.

Depending on the order in which the surfaces of the substrate are processed, the first orientation-changing device may be configured to change the orientation of the substrate by about 90 degrees or by about 180 degrees. The first orientation-changing device may be configured for changing the orientation of the substrate **100** from a substantially vertical orientation to a substantially horizontal orientation or from a substantially horizontal orientation to a substantially vertical orientation, as described herein. The first orientation-changing device may be configured for changing the orientation of the substrate **100** from a first substantially horizontal orientation to a second substantially horizontal orientation, as described herein.

Likewise, depending on the order in which the surfaces of the substrate are processed, the second orientation-changing device may be configured to change the orientation of the substrate by about 90 degrees or by about 180 degrees. The second orientation-changing device may be configured for changing the orientation of the substrate **100** from a substantially vertical orientation to a substantially horizontal orientation or from a substantially horizontal orientation to a substantially vertical orientation, as described herein. The second orientation-changing device may be configured for changing the orientation of the substrate **100** from a first substantially horizontal orientation to a second substantially horizontal orientation, as described herein.

At least one of the first orientation-changing device and the second orientation-changing device may be configured for changing the orientation of the substrate **100** from a substantially vertical orientation to a substantially horizontal orientation or from a substantially horizontal orientation to a substantially vertical orientation.

The first printing device, the second printing device and/or the third pattern deposition device may be a screen printing device.

A deposition apparatus **10** according to embodiments described herein may include a substrate support associated with the first printing device **1210**. The deposition apparatus **10** may include a substrate support associated with the second printing device **1230**. The deposition apparatus **10** may include a substrate support associated with the third printing device **1250**.

A deposition apparatus **10** according to embodiments described herein may include a vertically oriented substrate support. The term “vertically oriented substrate support” refers to a substrate support configured for supporting a substrate in a substantially vertical orientation, as described herein. At least one of the substrate support associated with the first printing device **1210**, the substrate support associated with the second printing device **1230** and the substrate

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support associated with the third printing device **1250** may be a vertically oriented substrate support.

The deposition apparatus **10** may include at least one horizontally oriented substrate support, e.g. one horizontally oriented substrate support or two horizontally oriented substrate supports. The term “horizontally oriented substrate support” refers to a substrate support configured for supporting a substrate in a substantially horizontal orientation, as described herein. At least one of the substrate support associated with the first printing device **1210**, the substrate support associated with the second printing device **1230** and the substrate support associated with the third printing device **1250** may be a horizontally oriented substrate support.

According to a further embodiment, a method of deposition on a substrate is provided. The method includes providing a substrate **100** in a substantially vertical orientation. The first substrate has a first main surface **120**, a second main surface **130** opposite the first main surface **120** and a side surface **110** between the first main surface **120** and the second main surface **130**. The method includes depositing a first conductive pattern **250** or a first resist mask **260** on the side surface **110** of the substrate **100** while the first substrate is in the substantially vertical orientation.

The first conductive pattern **250** and/or the first resist mask **260** may be deposited on the side surface **110** by a first deposition device **190** as described herein.

The substrate **100** may be provided, or supported, in the substantially vertical orientation by the first substrate support **180** as described herein.

Depositing the first conductive pattern **250** or the first resist mask **260** on the side surface **110** of the substrate **100** may include printing the first conductive pattern **250** or the first resist mask **260** on the side surface **110** of the substrate **100**.

A method according to embodiments described herein may include depositing, e.g. printing, a second conductive pattern **450** or a second resist mask **460** on the first main surface **120**, wherein a portion of the second conductive pattern **450** contacts a portion of the first conductive pattern **250** or a portion of the second resist mask **460** contacts a portion of the first resist mask **260**. The second conductive pattern **450** may be deposited by a second deposition device **330** as described herein. The second conductive pattern **450** may be deposited while the substrate **100** is held in a substantially horizontal orientation, e.g. by the second substrate support **320** as described herein.

A method according to embodiments described herein may include depositing, e.g. printing, a third conductive pattern **550** or a third resist mask **560** on the second main surface **130**, wherein a portion of the third conductive pattern **550** contacts a portion of the first conductive pattern **250** or a portion of the third resist mask **560** contacts a portion of the first resist mask **260**. The third conductive pattern **550** may be deposited on the second main surface **130** by a third deposition device **370** as described herein. The third conductive pattern **550** may be deposited while the substrate **100** is held in a substantially horizontal orientation, e.g. by the third substrate support **360** as described herein.

The method according to embodiments described herein may include changing the orientation of the substrate **100** from the substantially vertical orientation to a substantially horizontal orientation or from a substantially horizontal orientation to the substantially vertical orientation.

The method according to embodiments described herein may include changing the orientation of the substrate **100** from a first orientation to a second orientation. The method

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may include changing the orientation of the substrate **100** from the second orientation to a third orientation. At least one of the first orientation, the second orientation and the third orientation may be a substantially vertical orientation. At least one of the first orientation, the second orientation

The method according to embodiments described herein may include providing a stack of substrates **1010** in a substantially vertical orientation. The stack of substrates **1010** may include the substrate **100**. The stack of substrates **1010** in the substantially vertical orientation may have a top portion **1012**. The method may include depositing a conductive pattern **1150** or a resist mask **1160** on the top portion **1012** of the stack while the stack of substrates **1010** is supported in the substantially vertical orientation.

According to a further embodiment, a substrate structure **600** is provided. The substrate structure **600** includes a substrate **100** having a first surface and a second surface opposite the first surface. The first surface of the substrate **100** may be the first main surface **120** as described herein. The second surface may be the second main surface **130** as described herein. The substrate has a side surface **110** between the first surface and the second surface. The substrate structure **600** further includes a plurality of electrical connections between the first surface and the second surface. Each electrical connection includes a conductive line **452** on the first surface, a conductive line **252** on the side surface, and a conductive line **552** on the second surface. The conductive line **252** on the side surface contacts the conductive line **452** on the first surface and the conductive line **552** on the second surface.

According to a further embodiment, a substrate structure is provided. The substrate structure includes a substrate **100** having a first surface, a second surface opposite the first surface and a side surface **110** between the first surface and the second surface. The substrate structure includes a resist mask. The resist mask includes a first resist mask portion covering a portion of the side surface, the first resist mask portion having a first plurality of line-like openings. The first resist mask portion may be the first resist mask **260** as described herein. The resist mask includes a second resist mask portion covering a portion of the first surface, the second resist mask portion having a second plurality of line-like openings. The second resist mask portion may be the second resist mask **460** as described herein. The resist mask includes a third resist mask portion covering a portion of the second surface, the third resist mask portion having a third plurality of line-like openings. The third resist mask portion may be the third resist mask **560** as described herein. Each line-like opening **262** of the first plurality of line-like openings is adjacent to a line-like opening **462** of the second plurality of line-like openings and to a line-like opening **562** of the third plurality of line-like openings.

FIG. **13** shows a substrate support **1300** according to embodiments described herein.

According to a further embodiment, a substrate support **1300** is provided. The substrate support **1300** may be for supporting a substrate **100** in a substantially vertical orientation. The substrate support **1300** includes a first surface **1310**, the first surface **1310** being a stationary surface which is substantially vertical or a movable surface which is movable between a first position and a second position, wherein in the second position the first surface **1310** is substantially vertical. The substrate support **1300** includes a substrate attraction device **1320** adjacent to the first surface **1310**.

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The substrate **100** may have a first main surface **120**, a second main surface **130** and a side surface **110** as described herein. A substrate support **1300** according to embodiments described herein may be configured for holding the substrate **100** in a substantially vertical orientation in a manner such that the side surface **110** of the substrate **100** is exposed for deposition of a material on the side surface **110**, e.g. by the first deposition device **190** as described herein. In particular, no holding elements (such as clamps or the like) for holding the substrate **100** may be provided at or on the side surface **110**. In light thereof, the side surface **110** may be free for deposition of, e.g., a first conductive pattern **250** or a first resist mask **260** on the side surface **110**.

A substrate support **1300** according to embodiments described herein may include a support member **1330**. The first surface **1310** may be a surface, particularly a side surface, of the support member **1330**.

The substrate attraction device **1320** is configured to attract the substrate **100**. The substrate attraction device **1320** may be a substrate suctioning device. The substrate attraction device **1320** may be for suctioning the substrate **100** against the first surface **1310**. The substrate attraction device may be, for example, a vacuum chuck, an electronic chuck, or the like. The substrate attraction device **1320** and the substrate **100** may be arranged at opposite sides of the first surface **1310**.

The first surface **1310** may include a plurality of openings **1312**. The plurality of openings **1312** may allow air to pass through the first surface **1310**. The plurality of openings **1312** may allow the substrate attraction device **1320**, e.g. a substrate suctioning device, to suction the substrate **100** against the first surface **1310**. The substrate attraction device **1320** may be a vacuum suctioning device.

The substrate attraction device **1320** may be arranged in or at the support member **1330** of the substrate support **1300**.

The first surface **1310** may be a stationary surface, the stationary surface being a substantially vertical surface. The first surface **1310** may be fixed in a vertical orientation.

Alternatively, the first surface **1310** may be movable from a first position to a second position. In the first position, the first surface **1310** may be a substantially horizontal surface. In the second position, the first surface **1310** may be a substantially vertical surface.

FIGS. **14a-c** show a substrate support **1300** according to embodiments described herein. FIGS. **14a-b** show the first surface **1310** in the first position as described herein. FIG. **14c** shows the first surface **1310** in the second position as described herein.

As shown for example in FIGS. **14a-b**, the substrate **100** may be loaded onto the first surface **1310** while the first surface **1310** is in the first position. As shown in FIG. **14c**, the first surface **1310** may be moved from the first position to the second position by moving the first surface **1310** by an angle, e.g. an angle of about 90 degrees. When the first surface **1310** is in the second position, the substrate **100** is supported in a substantially vertical orientation. A first conductive pattern **250** or a first resist mask **260**, as described herein, may be deposited on the side surface **110** while the first surface **1310** is in the second position.

The substrate support **1300** may include an actuator for moving the first surface **1310** from the first position to the second position. For example, the first surface **1310** may be a surface of the support member **1330**, wherein the support member **1330** is movable over an angle, e.g. an angle of about 90 degrees, by the actuator.

Embodiments wherein the first surface **1310** is movable from a first position to a second position allow for receiving

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a horizontally oriented substrate, e.g. a substrate **100** on which in a previous stage a material was deposited while the substrate **100** was in a horizontal orientation. The substrate **100** can be received in the horizontal orientation and be rotated into a vertical orientation by the substrate support **1300**, so that a material can be deposited on the side surface **110** of the substrate **100**. Embodiments described herein thus facilitate switching between a horizontal orientation and a vertical orientation of the substrate **100**. Particularly, where the first surface **1310** is moved between the first position and the second position by an actuator, embodiments described herein allow for switching between a horizontal orientation and a vertical orientation of the substrate **100** in a fully automated manner.

A substrate support **1300** according to embodiments described herein may include a force member **1450** for pressing the substrate **100** against the first surface **1310**. FIG. **14c** shows an example of a force member **1450**.

For example, the force member **1450** may be a piston. The force member **1450** may be for holding the substrate **100** in the substantially vertical orientation. The force member **1450** may be configured to apply a force to the substrate, e.g. to the first main surface **120** or second main surface **130** as described herein. The force member may include a plate-like element for pressing against the substrate **100**. The force member may be configured for applying a substantially horizontal force to the substrate **100**. The force member **1450** may be movable in a substantially horizontal direction, as indicated by the arrow **1452** in FIG. **14c**. The force member **1450** and the substrate attraction device **1320** may be arranged on opposite sides of the substrate **100**.

A separator, e.g. a sheet of paper, may be arranged between the substrate **100** and the first surface **1310**.

A substrate support **1300** according to embodiments described herein may be configured for supporting a stack of substrates **1010**, as described herein, in a substantially vertical orientation. For example, a stack of substrates **1010** may be attracted against the first surface **1310** by the substrate attraction device **1320** and/or may be pressed against the first surface **1310** by the force member **1450**.

FIGS. **15a-b** shows a substrate support **1300** according to embodiments described herein. FIG. **15b** shows a top view.

According to a further embodiment, a substrate support **1300** for a deposition apparatus, particularly a printing apparatus, is provided. The substrate support **1300** may be for supporting a substrate **100** in a substantially vertical orientation. The substrate support **1300** includes a housing **1520** comprising a slit **1510** to receive the substrate **100** in the substantially vertical orientation. The slit **1510** may have a bottom surface. The slit **1510** may have a first surface **1310**, e.g. a surface of a first side wall of the slit. The slit **1510** may have a second surface **1512**, e.g. a surface of a second side wall of the slit. The second surface **1512** may face the first surface **1310**. The bottom surface may connect the first surface **1310** with the second surface **1512**. The first surface **1310** and the second surface **1512** may be substantially vertical surfaces. The bottom surface may be a substantially horizontal surface. The slit **1510** may be configured for receiving the substrate **100** between the first surface **1310** and the second surface **1512**.

The slit **1510** may be configured for receiving the substrate **100** in a manner such that the side surface **110** of the substrate **100** is exposed for depositing a material, e.g. a first conductive pattern or a first resist mask as described herein, on the side surface **110**.

The slit **1510** may have a top side. The housing **1520** may have a top side. The top side of the slit **1510** may be at the

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top side of the housing **1520**. The top side of the slit **1510** may be an open side for receiving the substrate **100**.

The slit **1510** may have a bottom side. The bottom side of the slit **1510** may be opposite the top side of the slit **1510**. The slit **1510** may include a substrate supporting surface **1550** for supporting the substrate **100** from below. The substrate supporting surface **1550** may be a substantially horizontal surface. The substrate supporting surface **1550** may be at the bottom side of the slit **1510**. The substrate supporting surface **1550** may be the bottom surface of the slit **1510**.

The slit **1510** may be configured for receiving a stack of substrates **1010**, as described herein, in a substantially vertical orientation. For example, a stack of substrates including two or three or more substrates can be inserted in the slit **1510**. The substrate supporting surface **1550** may be for supporting the stack of substrates **1010** in a substantially vertical orientation. FIG. **15b** shows a stack of two substrates received in the slit **1510**.

The substrate support **1300** may include a force member **1450** for holding the substrate **100** in the slit **1510**. FIG. **15b** shows an example of a force member **1450**. For example, the force member **1450** may be an arrangement including a plate pressing against the substrate **100** in the slit **1510**, possibly including one or more screws for adjusting the position of the plate.

The force member **1450** may be at least partially arranged in the slit **1510**. The force member **1450** may be configured to apply a force to the substrate **100**, e.g. to the first main surface **120** or second main surface **130** as described herein. The force member **1450** may include a plate-like element for pressing against the substrate **100**. The force member **1450** may be configured for applying a substantially horizontal force to the substrate **100**. The force member **1450** may be movable in a substantially horizontal direction, as indicated by the arrow **1580** in FIG. **15b**. The force member **1450** may be configured for pressing the substrate **100** against the first surface **1310** of the slit **1510**. A separator, e.g. a sheet of paper, may be arranged between the substrate **100** and the first surface **1310** of the slit **1510**.

The housing **1520** may surround the slit **1510**. The housing **1520** may, e.g., include or be made of aluminum. The housing **1520** may be shaped as a box.

The first substrate support **180** as described herein may be a substrate support **1300** according to embodiments described herein, e.g. a substrate support **1300** as shown in FIGS. **13**, **14a-c** or **15a-b**.

According to a further embodiment, a substrate support **1300** for supporting a substrate **100** is provided. The substrate support **1300** includes a first surface, the first surface being a stationary surface which is substantially vertical or a movable surface which is movable between a first position and a second position, wherein in the second position the first surface is substantially vertical. The substrate support **1300** includes a biasing device for biasing a substrate **100** against the first surface, the substrate **100** having a first main surface **120**, a second main surface **130** opposite the first main surface **120** and a side surface **110** between the first main surface **120** and the second main surface **130**. The substrate support **1300** may be configured for holding the substrate **100** in a substantially vertical orientation in a manner such that the side surface **110** of the substrate **100** is exposed for deposition of a material on the side surface **110**. The first surface may be the first surface **1310** as described herein, e.g. the first surface **1310** shown in FIG. **13** or **14a-c** or the first surface **1310** shown in FIG. **15b**. The biasing

device may include or be at least one of a substrate attraction device **1320** and a force member **1450** as described herein.

A substrate support **1300** according to embodiments described herein may be a substrate support for a deposition apparatus, particularly a printing apparatus. The substrate support may be included in the deposition apparatus. The deposition apparatus may include a deposition device, particularly a printing device. The deposition device may be for depositing a material on a substrate supported by the substrate support.

FIGS. **16a-b** shows a substrate support **1600** according to embodiments described herein.

According to a further embodiment, a substrate support **1600** for supporting a substrate **100** is provided. The substrate support **1600** includes a substrate receiving area **1610**. The substrate support **1600** includes a first recess **1630** at or near a first edge **1620** of the substrate receiving area **1610**.

The substrate support **1600** may be a substrate support for a deposition apparatus **10**, particularly a printing apparatus. The substrate support **1600** may be a nest, particularly a printing nest.

The substrate support **1600** may be configured for supporting the substrate **100** in a substantially horizontal orientation. The substrate receiving area **1610** may be a substantially horizontally oriented area. The second substrate support **320** and/or the third substrate support **360**, as described herein, may be a substrate support **1600** according to embodiments described herein.

The substrate receiving area **1610** may be configured for receiving the substrate **100**. The dimensions (e.g. length and width) of the substrate receiving area **1610** may correspond to the dimensions of the substrate **100** supported by the substrate support **1600**. The first edge **1620** of the substrate receiving area **1610** may correspond to, e.g. be aligned with, a first edge of the substrate **100** supported in the substrate receiving area **1610**.

The substrate support **1600** may have a substrate receiving surface **1650**. The substrate receiving area **1610** may be a region on the substrate receiving surface **1650**. The first recess **1630** may be a recess formed in the substrate receiving surface **1650**.

The first edge **1620** of the substrate receiving area **1610** may have a length in a first direction, e.g. direction **1602** shown in FIG. **16b**. The first recess **1630** may have a length in the first direction.

The first direction may be a horizontal direction. The first recess **1630** may be a strip-like recess having a length in the first direction. The length of the first recess **1630** in the first direction may be at least as long as, or longer than, the length of the first edge **1620** of the substrate receiving area **1610** in the first direction.

The first recess **1630** may have a width **1632** in a second direction, e.g. direction **1604** shown in FIGS. **16a-b**. The width **1632** of the first recess **1630** may be smaller than the length of the first recess **1630**. The second direction may be a horizontal direction. The second direction may be perpendicular to the first direction.

The first recess **1630** may have a width **1632** in the second direction from a first side of the first recess **1630** to a second side of the first recess **1630**. At least a portion of the first edge **1620** of the substrate receiving area **1610** may be located between the first side and the second side of the first recess **1630**.

A portion of the substrate receiving area **1610**, particularly an edge portion of the substrate receiving area **1610**, may overlap with the first recess **1630**, as shown for example in FIG. **16b**. The first side of the first recess **1630** may be

outside a periphery of the substrate receiving area **1610**. At least a portion of the second side of the first recess may be inside the periphery of the substrate receiving area.

The first recess **1630** may have a depth in a vertical direction.

The first recess **1630** may be configured for receiving deposition material **1680**. Particularly, the first recess **1630** may be configured for receiving deposition material **1680** resulting from depositing a layer or pattern **1690** on an edge region of the substrate **100** near the first edge **1620** of the substrate receiving area **1610**, such as e.g. deposition of a second/third conductive pattern or a second/third resist mask as described herein. By receiving the deposition material **1680** in the first recess **1630**, the deposition material **1680** does not accumulate on the substrate receiving surface **1650** of the substrate support **1600**, but is received in the first recess **1630**. In light thereof, the substrate receiving surface **1650** of the substrate support **1600** does not become dirty through an accumulation of deposition material thereon. Further, the presence of the first recess **1630** allows for usage of a screen with a bigger pattern for performing screen printing on the edge region of the substrate without making the substrate support dirty, since the deposition material urged through the screen at the periphery of the screen will be received in the first recess **1630**. Accordingly, an improved coverage of the edge region of the substrate **100** by the layer or pattern **1690** can be provided.

The substrate support **1600** according to embodiments described herein may include one or more further recesses like the first recess **1630**. For example, the substrate receiving area **1610** may have a second edge. The substrate support **1600** may have a second recess at or near the second edge. Like the first recess **1630**, the second recess may be configured to receive deposition material.

A substrate support **1600** according to embodiments described herein may be a substrate support for a deposition apparatus, particularly a printing apparatus. The substrate support may be included in the deposition apparatus. The deposition apparatus may include a deposition device, particularly a printing device. The deposition device may be for depositing a material on a substrate supported by the substrate support.

FIG. **17** shows a deposition apparatus **10** according to embodiments described herein.

According to a further embodiment, a deposition apparatus **10** is provided. The deposition apparatus **10** includes a first substrate support **180** for supporting a substrate **100**. The substrate has a first main surface **120**, a second main surface **130** opposite the first main surface **120** and a side surface **110** between the first main surface **120** and the second main surface **130**. The deposition apparatus **10** includes a first deposition device **190**. The first deposition device **190** may be for depositing a first conductive pattern **250** or a first resist mask **260** on the side surface **110** of the substrate **100**. The deposition apparatus **10** includes an observation device **1710**. The observation device may be for observing at least a portion of the first main surface **120**. The deposition apparatus **10** includes one or more alignment actuators **1720** connected to the observation device **1710**. The one or more alignment actuators **1720** may be configured to align the first deposition device **190** and the side surface **110** of the substrate **100** relative to each other.

The deposition apparatus **10** including the one or more alignment actuators **1720** may include any feature, or any feature combination, described herein within the context of other embodiments of the deposition apparatus **10**. For example, the deposition apparatus **10** may include a second

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deposition device 330, a third deposition device 370, a first orientation-changing device, a second orientation-changing device, and so on, or any combination thereof.

The first deposition device 190 may be arranged for depositing the first conductive pattern 250 or the first resist mask 260 on the side surface 110 of the substrate 100 while the substrate 100 is supported by the first substrate support 180. The observation device 1710 may be configured for observing the at least a portion of the first main surface 120 while the substrate 100 is supported by the first substrate support 180. The one or more alignment actuators 1720 may be configured for aligning the first deposition device 190 and the side surface 110 relative to each other while the substrate 100 is supported by the first substrate support 180. The first substrate support 180 may be configured to support the substrate 100 in a substantially vertical orientation.

That the one or more alignment actuators 1720 are connected to the observation device 1710 can be understood in the sense that the observation device 1710 is in communication with the one or more alignment actuators 1720. The one or more alignment actuators 1720 may be connected to the observation device 1710 in a manner such that data obtained by observing the first main surface 120 of the substrate 100 can be provided from the observation device 1710 to the one or more alignment actuators 1720, possibly via a controller as described herein.

That the first deposition device 190 and the side surface 110 of the substrate 100 are aligned relative to each other does not necessarily mean that the entire first deposition device 190 and the side surface 110 are aligned relative to each other. For example, a portion of the first deposition device 190, such as e.g. a deposition head or printing head, may be displaced to align the portion relative to the side surface 110, while a remaining portion of the first deposition device 190 may remain stationary.

In some implementations, the one or more alignment actuators 1720 may be part of the first deposition device 190. Alternatively, the one or more alignment actuators 1720 and the first deposition device 190 may be different components of the deposition apparatus. The first deposition device 190 may be mounted to the one or more alignment actuators 1720 or vice versa. A portion of the first deposition device 190, e.g. a deposition head or printing head, may be mounted to the one or more alignment actuators 1720, or vice versa.

In some implementations, the one or more alignment actuators 1720 may be part of the first substrate support 180, e.g. a printing nest. Alternatively, the one or more alignment actuators 1720 and the first substrate support 180 may be different components of the deposition apparatus. The first substrate support 180 may be mounted to the one or more alignment actuators 1720 or vice versa.

The observation device 1710 may be configured to provide data obtained by observing at least a portion of the first main surface 120 to the one or more alignment actuators 1720. The data may be data regarding the position of the at least a portion of the first main surface 120. The one or more alignment actuators 1720 may be configured to align the first deposition device 190 and the side surface 110 of the substrate 100 relative to each other based at least on the data.

The first deposition device 190 may be configured for depositing the first conductive pattern 250 or the first resist mask 260 on the side surface 110 of the substrate 100 after the first deposition device 190 and the side surface 110 of the substrate have been aligned with respect to each other by the one or more alignment actuators 1720.

The first substrate support 180 may be configured for supporting the substrate 100 in a substantially vertical

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orientation. The first deposition device 190 may be configured for depositing the first conductive pattern 250 or the first resist mask 260 on the side surface 110 of the substrate 100 while the substrate 100 is supported in the substantially vertical orientation by the first substrate support 180. The observation device 1710 may be configured for observing the at least a portion of the first main surface 120 while the substrate 100 is supported in the substantially vertical orientation by the first substrate support 180.

The substrate 100 may have an alignment mark 1810 on the first main surface 120, as shown for example in FIG. 18. The observation device 1710 may be configured to observe the alignment mark 1810. The observation device 1710 may be configured to provide data obtained by observing the alignment mark 1810 to the one or more alignment actuators 1720. The data may be data regarding the position of the alignment mark 1810. The one or more alignment actuators 1720 may be configured to align the first deposition device 190 and the side surface 110 of the substrate relative to each other based at least on the data.

The first main surface 120 may include a plurality of alignment marks. For example, in the case of a rectangular substrate, an alignment mark can be provided in the two upper corners of the first main surface 120. The observation device 1710 may be configured for observing the plurality of alignment marks. The one or more alignment actuators 1720 may be configured for aligning the side surface 110 and the first deposition device 190 relative to each other at least based on data obtained by observing the plurality of alignment marks.

The alignment mark(s) on the first main surface 120 of the substrate 100 may be already available, e.g. pre-formed, for aligning the first main surface 120, e.g., for a deposition process in which a material is deposited on the first main surface 120. Embodiments described herein make use of the alignment mark(s) on the first main surface 120 to align a different surface of the substrate 100, namely the side surface 110, with respect to a deposition device for depositing a material on the side surface 110.

The observation device 1710 may be an inspection device for inspecting at least a portion of the first main surface 120 of the substrate 100. The observation device 1710 may be or include an imaging device, particularly an optical imaging device such as e.g. a camera, configured to make an image of the at least a portion of the first main surface 120. The observation device 1710 may be configured to make an image of the alignment mark 1810.

The alignment performed by the one or more alignment actuators 1720 as described herein is a relative alignment of the first deposition device 190 and the side surface 110 of the substrate 100 with respect to each other. The one or more alignment actuators 1720 may be configured for moving the first deposition device 190 (as illustrated e.g. in FIG. 17), for moving the substrate 100 or for moving both the first deposition device 190 and the substrate 100 to align the first deposition device 190 and the side surface 110 of the substrate relative to each other.

The one or more alignment actuators 1720 may be configured for providing a translational alignment and/or an angular alignment of the first deposition device 190 and the side surface 110 of the substrate relative to each other.

The one or more alignment actuators 1720 may be configured for translating the first deposition device 190, or at least a portion thereof, such as e.g. a deposition head or printing head of the first deposition device 190. The first deposition device 190 or the portion thereof may be moved, e.g. translated, by the one or more alignment actuators 1720

in a substantially vertical direction and/or in a substantially horizontal direction. Additionally or alternatively, the one or more alignment actuators **1720** may be configured for changing an angle of the first deposition device **190**, or at least a portion thereof, such as a deposition head or printing head. The angle may be an angle with respect to a substantially horizontal axis.

The one or more alignment actuators **1720** may be configured for translating the substrate **100**. The substrate **100** may be translated in a substantially vertical direction and/or in a substantially horizontal direction. The one or more alignment actuators **1720** may be configured for changing an angle of the substrate **100**. The angle may be an angle with respect to a substantially horizontal axis.

A deposition apparatus **10** according to embodiments described herein may include a controller. The controller may be connected to the observation device **1710**. Additionally or alternatively, the controller may be connected to the one or more alignment actuators **1720**. The controller may be configured to determine a target relative position for the first deposition device **190** with respect to the side surface **110** of the substrate **100** based on data provided by the observation device **1710**. The data may be data regarding the position of the at least a portion of the first main surface **120** or data regarding the position of the alignment mark **1810**. The controller may be configured for issuing an instruction to the one or more alignment actuators **1720**. The controller may be configured to instruct the one or more alignment actuators to move the first deposition device **190**, or a portion thereof, and/or to move the substrate **100**, to provide the first deposition device **190** and the substrate **100** in the target relative position. In the target relative position, the first deposition device **190** and the substrate **100** are well-aligned relative to each other.

FIG. **19** shows a deposition apparatus **10** according to embodiments described herein.

The observation device **1710**, e.g. a camera, may be connected to the first deposition device **190**. The observation device may be mounted to the first deposition device **190**, as shown for example in FIG. **19**. For example, the observation device **1710** may be connected or mounted to a deposition head, e.g. a printing head, of the first deposition device **190**.

A deposition apparatus according to embodiments described herein may include a reflective surface **1912**. The reflective surface **1912** may be a reflective surface of a reflective member **1910**, e.g. a mirror. The observation device **1710**, e.g. a camera, may be configured for observing at least a portion of the first main surface **120** via the reflective surface **1912**. The observation device **1710** may be arranged at an elevated position with respect to the reflective surface **1912**. For example, the observation device **1710** may be attached or mounted to the first deposition device **190**, particularly to a printing head of the first deposition device **190**. The reflective surface **1912** may face the substrate **100** supported by the first substrate support **180**. In particular, the reflective surface **1912** may face the vertically oriented substrate **100**. The reflective surface **1912** may be arranged to reflect light emanating from a region near the first main surface **120** to the observation device **1710**, as e.g. indicated in FIG. **19** by the dashed lines.

A deposition apparatus according to embodiments described herein may include a light source **1920**. The light source **1920** may be for illuminating the first main surface **120**. The light source **1920** may be arranged in a region near a substrate receiving area of the first substrate support **180** or near the substrate **100** supported by the first substrate support **180**. The light source **1920** and the reflective surface

1912 may be arranged on the same side of the substrate **100** or substrate receiving area (as shown e.g. in FIG. **19**), or on different sides of the substrate **100** or substrate receiving area (e.g. if the substrate **100** is transparent). Light from the light source **1920** may travel to the reflective surface **1912** and be reflected by the reflective surface to the observation device. The observation device **1710** may be configured for observing at least a portion of the first main surface **120** via the reflective surface **1912**.

Whereas some embodiments described herein use a light source **1920** and a reflective surface **1912** to facilitate the observation of the substrate **100** by the observation device **1710**, neither the light source **1920** nor the reflective surface **1912** is necessary for observing the substrate **100**. For example, the observation device **1710** may be arranged to face the substrate **100** supported by the first substrate support **180**. In such a set-up, a reflective surface **1912** is not needed. Further, depending on the lighting conditions of the deposition apparatus **10**, a light source **1920** for illuminating the first main surface **120** may not be necessary.

In the above description of the deposition apparatus **10**, the observation device **1710** is arranged for observing at least a portion of the first main surface **120**, as shown for example in FIG. **17**. Alternatively, the observation device **1710** can be arranged for observing at least a portion of the second main surface **130**.

FIGS. **20a-b** show a deposition apparatus **10** according to embodiments described herein.

A deposition apparatus **10** according to embodiments described herein may include one or more inspection units **2010** for determining an alignment of the side surface **110** with respect to a substantially horizontal direction, e.g. direction **2030** shown in FIGS. **20a-b**. The one or more inspection units **2010** can be combined with any embodiment of a deposition apparatus **10** as described herein.

The one or more inspection units **2010** may be arranged above a substrate receiving area of the first substrate support **180** or above the substrate **100** supported by the first substrate support **180**, particularly a vertically oriented substrate or substrate receiving area. The one or more inspection units **2010** may be two or more inspection units, e.g. two cameras or distance sensors. The two or more inspection units may include a first inspection unit and a second inspection unit spaced apart from each other by a horizontal distance, as shown for example in FIGS. **20a-b**.

The one or more inspection units **2010** may be configured to inspect the side surface **110** of the substrate **100** supported by the first substrate support **180**. The one or more inspection units **2010** may be configured to inspect an upper edge **2020** of the substrate **100** supported by the first substrate support **180**. The upper edge **2020** may be an edge of the side surface **110**. The upper edge **2020** may be a substantially horizontal edge of the substrate **100** supported by the first substrate support **180**.

The one or more inspection units **2010** may be part of the observation device **1710** as described herein, may be the observation device **1710**, or may be separate from the observation device **1710**.

The one or more inspection units **2010** may be configured for determining a height of the substrate **100**, particularly a vertically oriented substrate, supported by the first substrate support **180**. In light thereof, embodiments described herein allow for dynamically changing the snap-off according to the height of the substrate **100** and the difference of substrate size can be compensated for on a substrate-by-substrate basis.

Additionally or alternatively, the one or more inspection units may be for determining a horizontal alignment of the upper edge **2020**, particularly for determining whether the upper edge **2020** is aligned with a substantially horizontal direction, e.g. direction **2030** shown in FIGS. **20a-b**. The one or more inspection units may be for determining whether an angle exists between the upper edge **2020** and the substantially horizontal direction. If no angle or an angle below a small threshold angle exists, the upper edge **2020** may be well aligned, as shown for example in FIG. **20a**. If an angle above the threshold value exists, the upper edge **2020** may not be well aligned, as shown for example in FIG. **20b**. The angle between the upper edge **2020** and the direction **2030** is drawn in an exaggerated manner in FIG. **20b**.

By having the one or more inspection units **2010**, embodiments described herein provide the advantage that the deposition process, e.g. a screen printing process, can be interrupted or stopped if the tilting angle of the substrate **100**, i.e. the angle between the upper edge **2020** and the substantially horizontal direction **2030**, is too large. In light thereof, damage to e.g. the screen of the first deposition device, which can arise if the tilting angle of the substrate is too large, can be avoided.

According to a further embodiment, a method of alignment is provided. The method includes providing a substrate **100** having a first main surface **120**, a second main surface **130** opposite the first main surface **120** and a side surface **110** between the first main surface **120** and the second main surface **130**. The method includes aligning a first deposition device **190** and the side surface **110** of the substrate relative to each other, the aligning being based at least on data obtained by observing at least a portion of the first main surface **120**.

The aligning of the first deposition device **190** and the side surface **110** relative to each other may be performed by one or more alignment actuators **1720** as described herein.

The method may include depositing a first conductive pattern **250** or a first resist mask **260**, as described herein, on the side surface **110** of the substrate **100** using the first deposition device **190**. The first conductive pattern **250** or the first resist mask **260** may be deposited on the side surface **110** after the first deposition device **190** and the side surface **110** of the substrate have been aligned with respect to each other based on the data obtained by observing the first main surface **120**.

The method may include supporting the substrate **100** in a substantially vertical orientation, e.g. using a first substrate support **180** as described herein. The method may include observing the at least a portion of the first main surface **120** while the substrate **100** is supported in the substantially vertical orientation. The first deposition device **190** and the side surface **110** of the substrate **100** may be aligned relative to each other while the substrate **100** is supported in the substantially vertical orientation. The first conductive pattern **250** or the first resist mask **260** may be deposited on the side surface **110** of the substrate **100** while the substrate **100** is supported in the substantially vertical orientation.

The substrate **100** may have an alignment mark **1810**, as described herein, on the first main surface **120**. Particularly, the at least a portion of the first main surface **120** may include the alignment mark **1810**. The method may include observing the alignment mark **1810**, particularly observing a position of the alignment mark **1810**. The data obtained by observing at least a portion of the first main surface **120** may be data obtained by observing the alignment mark **1810**. The method may include aligning the first deposition device **190**

and the side surface **110** of the substrate relative to each other based at least on the data obtained by observing the alignment mark **1810**.

Observing at least a portion of the first main surface **120** or observing the alignment mark **1810** may include making one or more images of the at least a portion or of the alignment mark **1810**, e.g. using a camera.

The method may include determining an alignment of the side surface **110** with respect to a substantially horizontal direction, as described herein. The alignment may be determined using one or more inspection units **2010** as described herein.

While the foregoing is directed to embodiments of the disclosure, other and further embodiments of the disclosure may be devised without departing from the basic scope thereof, and the scope thereof is determined by the claims that follow.

The invention claimed is:

1. A deposition apparatus, comprising:

a first substrate support for supporting a substrate in a substantially vertical orientation, the substrate having a first main surface, a second main surface opposite the first main surface and a side surface between the first main surface and the second main surface; and

a first deposition device for depositing a first conductive pattern or a first resist mask on the side surface of the substrate while the substrate is supported in the substantially vertical orientation by the first substrate support.

2. The deposition apparatus according to claim 1,

wherein the first substrate support is configured for supporting a stack of substrates in a substantially vertical orientation, the stack of substrates in the substantially vertical orientation having a top portion,

wherein the first deposition device is configured for depositing a conductive pattern or a resist mask on the top portion of the stack while the stack is supported in the substantially vertical orientation by the first substrate support.

3. A method of deposition on a substrate, comprising:

providing a substrate in a substantially vertical orientation, the substrate having a first main surface, a second main surface opposite the first main surface and a side surface between the first main surface and the second main surface, the side surface being a surface at a top portion of the substrate and having a width corresponding to a thickness of the substrate; and

printing a first conductive pattern or a first resist mask on the side surface of the substrate while the substrate is in the substantially vertical orientation.

4. The method according to claim 3, further comprising at least one of:

depositing a second conductive pattern or a second resist mask on the first main surface, wherein a portion of the second conductive pattern contacts a portion of the first conductive pattern or a portion of the second resist mask contacts a portion of the first resist mask; and

depositing a third conductive pattern or a third resist mask on the second main surface, wherein a portion of the third conductive pattern contacts a portion of the first conductive pattern or a portion of the third resist mask contacts a portion of the first resist mask.

5. The method according to claim 4, wherein:

the first conductive pattern includes a first plurality of conductive lines, the second conductive pattern includes a second plurality of conductive lines, and the third conductive pattern includes a third plurality of

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conductive lines, each conductive line of the first plurality of conductive lines contacting a conductive line of the second plurality of conductive lines and a conductive line of the third plurality of conductive lines to form an electrical connection from the first main surface to the second main surface, or

the first resist mask includes a first plurality of line-like openings, the second resist mask includes a second plurality of line-like openings, and the third resist mask includes a third plurality of line-like openings, each line-like opening of the first plurality of line-like openings being adjacent to a line-like opening of the second plurality of line-like openings and to a line-like opening of the third plurality of line-like openings.

6. The method of claim 3, wherein the substrate is provided in the substantially vertical orientation by a first substrate support and the first conductive pattern or the first resist mask is deposited on the side surface of the substrate by a first deposition device, the method further comprising:

supporting the substrate in a substantially horizontal orientation by a second substrate support; and

depositing a second conductive pattern or a second resist mask on the first main surface of the substrate by a

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second deposition device while the substrate is supported by the second substrate support.

7. The method of claim 6, further comprising:

supporting the substrate in a substantially horizontal orientation by a third substrate support; and

depositing a third conductive pattern or a third resist mask on the second main surface of the substrate by a third deposition device while the substrate is supported by the third substrate support.

8. The method of claim 3, further comprising:

changing the orientation of the substrate from a first orientation of the substrate to a second orientation of the substrate.

9. The method of claim 3, further comprising:

supporting a stack of substrates in a substantially vertical orientation, the stack of substrates including the substrate, the stack of substrates in the substantially vertical orientation having a top portion; and

depositing a conductive pattern or a resist mask on the top portion of the stack while the stack is supported in the substantially vertical orientation.

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